# PROGRESS REPORT III

项目进度报告（三）

Tsinghua University, Beijing, 100084

University of Science and Technology Beijing, 100083

Dec, 2015

# ABSTRACT

With the fast development of elevator industry, passengers have been paying closer attention to the riding comfort than the function itself. Since millions of elevators are in running and it is challenging for manufacturers to provide enough professional maintenance technicians, a modern sensor network together with condition monitoring & diagnosis system is in urgent need.

This project aims to study the health condition and riding comfort of elevator system from the perspective of vibration. In the first phase of this project, both vertical and horizontal vibration model were established to study the vibration characteristics of elevator system and solve its dynamic response under various conditions. Further in phase 2, a series of lab experimental as well as in–situ tests were carried out, based on which specific vibration measurement scenarios, typical fault diagnosis methods & indicators were proposed. Finally in phase 3, specific on-site tests including baseline, rope wear, rope loose and rope local fault were conducted to validate the feasibility and effectiveness of proposed methods/indicators. Main achievements in phase 3 can be summarized as:

1. Test scenarios for baseline and typical faulty case experiment;
2. On-site test dataset in Jinan pilot elevator;
3. Validation of the correctness and effectiveness of former proposed frequency domain indicators in real elevator vibration fault diagnosis;
4. Equipment specifications and instructions for elevator vibration signal analysis.

The rest of this report is structured as follows. The objects and background of research in phase 3 are respectively given in Section 1 and 2. In Section 3, the effectiveness of our formerly proposed rope passing frequency indicator in diagnosing rope wear fault is evaluated using on-site elevator vibration datasets. Similarly, the performance of detecting rope end loose fault using rope swing frequency and natural frequency is evaluated as shown in Section 4. The identification of rope local defect by rope swing frequency is also presented in Section 5. Then in Section 6 we exhibit the equipment specification and basic knowledge of elevator vibration signal analysis. Finally the conclusion is summarized in Section 7.

**Key words:** Elevator, fault diagnosis, indicator, signal processing

# TABLE OF CONTENTS

**ABSTRACT** ..... I
**TABLE OF CONTENTS** ..... III
**1 OBJECTS** ..... 5
**2 RESEARCH BACKGROUND** ..... 7
**3 ON-SITE VIBRATION TEST** ..... 11
3.1 Measuring points ..... 11
3.1.1 Traction motor ..... 11
3.1.2 Car frame ..... 12
3.1.3 Counterweight ..... 13
3.1.4 Pulleys ..... 13
3.2 Rope fault simulation ..... 14
3.2.1 Baseline ..... 14
3.2.2 Rope wear/fatigue ..... 15
3.2.3 Rope end loose ..... 15
3.2.4 Rope local fault ..... 16
3.3 Test scenario ..... 17
3.3.1 Vertical modal test ..... 17
3.3.2 Rope swing test ..... 17
3.3.3 Car travel test ..... 18
**4 APPLICATION 1: DETECTION OF ROPE WEAR** ..... 19
4.1 Rope passing frequency ..... 19
4.2 Baseline ..... 20
4.3 Rope wear detection ..... 22
4.4 Conclusion ..... 24
**5 APPLICATION 2: DETECTION OF ROPE LOCAL FAULT** ..... 25
5.1 Rope swing frequency ..... 25
5.2 Baseline ..... 26
5.3 Rope local fault detection ..... 28
5.4 Conclusion ..... 29
**6 APPLICATION 3: DETECTION OF ROPE LOOSE** ..... 31
6.1 System natural frequency & rope swing frequency ..... 31
6.2 Baseline ..... 32

6.3 Rope loose detection ..... 34
6.4 Conclusion ..... 36
**7 FAQS ON INSTRUMENT AND MEASUREMENT** ..... 37
7.1 Sensors ..... 37
7.2 Data acquisition system ..... 40
**8 INTRODUCTION OF TIME-FREQUENCY ANALYSIS IN ELEVATOR MONITORING & DIAGNOSIS** ..... 43
8.1 Linear time-frequency representations ..... 43
8.1.1 Short-time Fourier transform ..... 43
8.1.2 Wavelet transform ..... 44
8.2 Bilinear Time–Frequency Distribution ..... 45
8.3 Applications ..... 46
**9 CONCLUSIONS** ..... 49
**REFERENCE** ..... 51

# 1 OBJECTS

Elevators are more and more commonly equipped in buildings all around the world, and its safety as well as riding comfort is an important issue all manufacturers will have to face. However, most of current literatures on elevators only focus on its vibration responses or simply periodical inspection instructions, while deep research into its fault mechanisms and specific monitoring/diagnosing methods or indicators are very limited.

In this project both parties aim to collaborate in monitoring elevator riding comfort and diagnosing potential fault by vibration measurement and analysis. Three main phases composed of this project. In phase 1 the vibration properties of elevator system was studied based on 17 DOF vertical vibration model and 9 DOF horizontal vibration model. Phase 2 focus on the vibration measurement and diagnosis method for elevator system, especially the motor, horizontal guide rail and most of all, the ropes. Main tasks of phase 3 include:

Task 1: Specific for rope tension signals, propose indicators for reflecting the force difference between each hoist rope and monitoring the running condition of traction system. The frequency structure of the rope tension signals will also be studied for fault feature extraction and revealing inner link between car vibration and hoist rope tension.

Task 2: Specific for car vibration signals, propose indicators for vibration magnitude classification and monitoring general ride quality. Vibration signals in horizontal directions will be used for guiding parts monitoring while vertical vibration signals will contribute to vertical smoothness monitoring. Combination of vibration signals in three directions will also be studied for environment estimation under extreme conditions.

Task 3: Study the structure/content feature of the object elevator's running signals and propose best suited analysis methodologies as well as procedures for the objective elevator monitoring, based on comparison of existing analysis approaches like wavelet transform and experience of the previous study and experiment.

Task 4: Detailed condition evaluation report for the objective elevator, using all former proposals of vibration model, experimental data, specific indicators for car vibration as well as hoist rope tension, preprocessing algorithms, specific analysis methodologies and proposed procedures. The fusion of these knowledge will form a condition evaluation report from an overall perspective, which can be used for maintenance/design guidance and application extension in the future.

# 2 RESEARCH BACKGROUND

Combined with contents in former reports on elevator modeling and vibration measurement & analysis, in this section the research background and recent advances of elevator modeling, monitoring and fault diagnosis are briefly summarized as follows.

**(1) Elevator modeling**

Elevator vertical vibration, which is stimulated by motor output torque and transferred mainly through hoist rope, have been generally under focus. Zhang and Mei [1] build a 10 DOF vertical vibration model based on Matlab to carry out vibration feature calculation. Zhen [2] etc., constructs an simple 1:1 elevator vertical and horizontal model, and further developed a dynamic analyzing software based on C programming language. A Mitsubishi elevator is also used for effectiveness evaluation. Wang [3] etc., studied 2:1 elevator trying to extract the frequency characteristic and inner relationship, with simulated excitation as well. Chen and Yu [4] focus on a TRJT1000/1.6-JXPC elevator in order to build a vertical 7 DOF vibration model. They also proposed an optimization method for kinetic characteristics analysis, together with Fortran program. However, they didn't take the time-variant parameters into consideration. Lei [5] studied the lab elevator and build a simulation model trying to adjust the controlling system for kinetic characteristics improvement. Gang [6] takes the mass of hoist rope and compensation rope into consideration, and use genetic algorithm to optimize the dynamic behavior based on elevator sensitivity. Zhang and Yang [7-8] discretize a 1:1 elevator 7DOF model and transform the time variant model into series of transient systems for obtaining numerical solution. Nai [9-10] study the key point of time-varying parameters modeling and build a 10DOF vertical model, in which the mass of hoist rope are discretized.

Research on horizontal vibration analysis are relatively less, because the force transmission path, horizontal structure and parameters are all related to the vibration response and are of complex mechanism. Feng and Zhang [11] systematically study the active control of high speed elevator horizontal vibration and proposed an active guide shoe based on hydraulic control. Both simulation and on-site test are applied to validate the effectiveness of active guide shoe and controlling strategy. Robust control rules are also applied to reduce the horizontal vibration. Fu [12] regards the connection between car frame and car as rigid, and further build a horizontal 5DOF model using transformation of

coordinates. Different guide rail excitations are added to obtain possible responses. But the horizontal damping cannot be ignored in real world case. Zhang [13] studied the influence of guide rail unevenness and running speed on horizontal vibration using a 12 DOF nonlinear model. Zhu [14] established a stochastic volatility model with fixed/kinetic hoist rope and corresponding vibration equations and boundary values, which lays a foundation for the study of relationship between hoist rope and car vibration.

From aforementioned literature, we can find that although vast amounts of work have been done in elevator vibration analysis field, most of them stay on the theoretical simulation stage and the factors having been taken into consideration are not complete. In vertical vibration analysis, the mass of hoist rope ought to be added and they are time variant. Guide rail excitation in horizontal direction always features randomness and are sensitive to fault. The elevator vertical and horizontal vibration demand for deep and complete research, together with more accurate model and on-site test.

**(2) Instrument and measurement on elevator monitoring**

The evolution of equipment maintenance mode has experienced three main phases. In 1950s the equipment are maintained in breakdown maintenance (BM) mode, while equipment are repaired only after breakdown. It caused huge property and life loss. Then in 1960s, preventive maintenance (PM) mode was proposed to plan for regular maintenance. The drawback of preventive maintenance, maintenance resources waste, had been recognized in 1980s and condition based maintenance (CBM) were proposed. Thereafter, measurement and fault diagnosis techniques have been rapidly developing. Nowadays a derivative of condition based maintenance termed reliability centered maintenance (RCM) is popular all around the world, which focus on the readability of equipment and proactive maintenance. For elevator system, the condition based maintenance and reliability centered maintenance are both developing and condition monitoring and fault diagnosis for elevator are still under hot research.

Elevator measurement techniques are developing rapidly in these years. Acceleration characteristics is a key index for elevator ride quality and acceleration measurement has evolved from the early WR7300F recorder by GRAPHETEC Inc to micro and digital measurement system like DT-2 elevator acceleration tester [15]. Compound vibration measurement system like ADIAS and EVA-625 have also been wide applied in car vibration and noise measurement. They can further recognize the acceleration, velocity, displacement and further carry out simple analysis. Besides, vibration measurement for traction motor has also been under study and a 911KT traction motor vibration tester has been developed and used in motor fault diagnosis [16]. Guide rail relates much to the system horizontal vibration and Tianjin University developed a guide rail detecting system for guide rail measurement [17]. Elevator balance coefficient is one of the parameters

which ensure safety, and balance coefficient measurement based on displacement transducer, current and voltage transducer have been developed [18].

**(3) Condition monitoring and fault diagnosis method/systems**

Generally elevator monitoring and fault diagnosis composed of two parts: monitoring the running of system, as well as diagnosing detail fault once abnormal signal is detected. Elevator system monitoring and diagnosis has developed from coarse diagnosis to precise and intelligent diagnosis, and from single model diagnosis to network diagnosis. Two categories of monitoring and diagnosis methods can be introduced as below [19]:

a. Mathematical model based methods
   The key of model-based monitoring and diagnosis methods lies in the prediction of monitored quantities [20], which shall be compared with true ones. Theoretically it is perfect since it systematically reveals the vibration mechanisms and all possibilities. Yet complex real-world interferences and errors have to be taken into consideration, and it can only be fulfilled after most of challenges have been concurred.

b. Intelligent methods
   Except for ideal vibration model based methods, a series of other solutions were proposed, including expert system, fault tree-based methods, and neural network-based methods [21-24]. These methods mainly take the whole elevator system as a series of components and try to build logical solutions to understand potential faults like door failures and control system faults. However, these conclusions mostly based on inference and has to be validated and improved by massive tests and experience.

Monitoring and diagnosing methods based on vibration measurement and specific fault mechanisms have also been in development owing to more powerful measurement equipment and algorithms. Special fault detection techniques like ultrasonic flaw detection and laser collimator [25] have been applied in elevator fault diagnosis field. Nevertheless, the research achievements on elevator machinery fault diagnosis have been very limited. Since the theory of vibration based analysis are widely studied, the elevator fault diagnosis method mostly fall in this field. A few work on elevator fault diagnosis technique have been reported [26-31]. However, specific fault diagnosis method or indicators are very limited [32-35], and most diagnosis method doesn’t have innovative breakthrough in fault characteristics and inherent indicators.

Intelligent & networked monitoring and control system has been applied to elevators

in most big elevator companies [36-38]. It helps record elevator health information remotely and quick react to accident. However, current monitoring systems mostly focus on basic running parameters like acceleration, velocity and noise level, while mechanical characteristics which are related to fault evolution still demand for deep research.

# 3 ON-SITE VIBRATION TEST

A KONE PTW13/10-19 elevator (30075851) is chosen as the project pilot elevator. It is installed in a 4-storey building and has been running for 9 years. Seizing the chance of replacing old ropes, the research team managed to carry out a series of experiments in both baseline and faulty cases. In this section, the detail of vibration measurement and on-site test scenario are exhibited.

## 3.1 Measuring points

According to former experience, it is necessary to collect vibration signals from car frame, car pulley, traction motor and the counterweight, because these signals contain massive health information and can be conveniently perceived by accelerometers. Similar to the on-site test in Beijing, we used 3 sets of data acquisition systems (equipment specifications can be found in Section 7) to cover these measuring points.

### 3.1.1 Traction motor

The traction motor is one of the core component of elevator system, because it works as the power source and has direct contact with hoist rope. Thus the vibration signal perceived from traction motor can be used for elevator system monitoring and fault diagnosis. Yet it has to be noted that interferences from current fluctuating and electromagnetic torque ripple are quite prominent, and useful characteristic frequency components may not be dominant.

Since the traction motor of pilot elevator is fixed on roof, we employed a set of wired vibration data acquisition system and 4 accelerometers in total were mounted on the motor case, as presented in Figure 3.1.

Fig 3.1 Motor vibration measuring points

### 3.1.2 Car frame

All former experiments and analysis have indicate that car frame vibration signal is significant for indirectly reflecting the property of hoist rope system. We measured the car frame vibration with 3 unidirectional accelerometers, as can be seen in Figure 3.2.

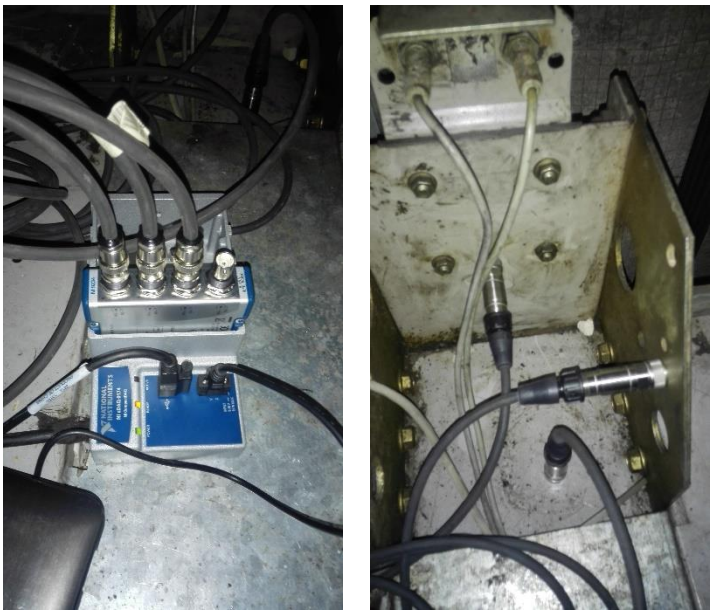


Fig 3.2 Car frame vibration measuring points

### 3.1.3 Counterweight

For comparison with car frame vibration, counterweight vibration in vertical direction was also measured using a wireless accelerometer, see Figure 3.3.

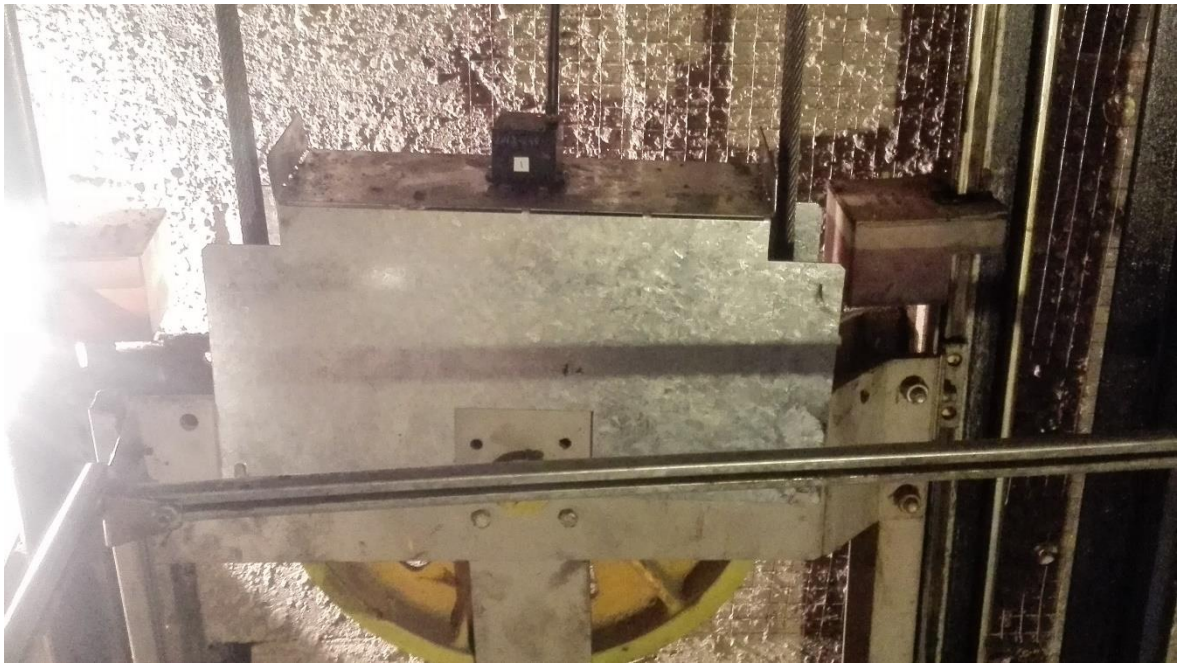


Fig 3.3 Counterweight vibration measuring points

### 3.1.4 Pulleys

As one of the main innovation point, we chose to measure the vibration of car frame pulleys since they are in direct contact with hoist ropes, which go through the whole elevator system and is rich in health information. Considering the fact that pulley whose axis is fixed to car frame is not always within reach, 2 wireless accelerometers were mounted respectively in radial and axial direction, as exhibited in Figure 3.4.

Fig 3.4 Pulley vibration measuring points

## 3.2 Rope fault simulation

In order to validate the effectiveness of proposed frequency domain indicators in detecting rope fault, we introduced typical failure types including rope wear/fatigue, rope end loose and rope local fault and collected vibration dataset under both baseline and faulty conditions for comparison.

### 3.2.1 Baseline

Pilot elevator equipped with brand new hoist ropes (as shown in Figure 3.5) is provisionally considered as healthy one, i.e., the baseline.

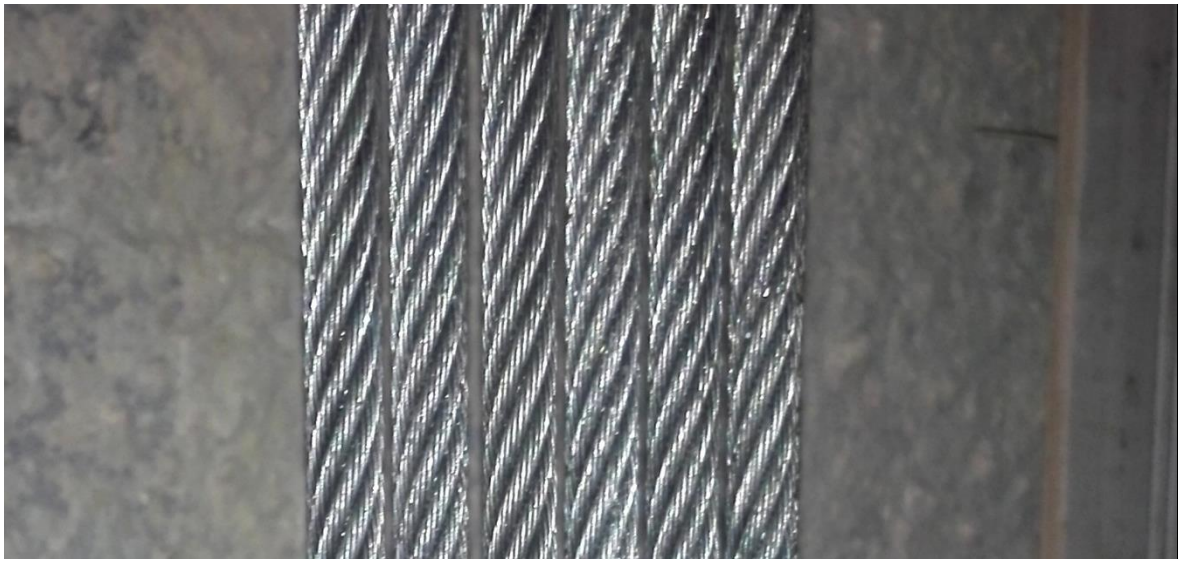


Fig 3.5 Healthy rope

Six hoist ropes, each of which is 8mm thick and has lay length of 8*6mm, were used in pilot elevator. We also measured the tension force of each rope and consider the case with balanced force distribution as healthy one. The measured rope tension forces are 117, 126, 106, 129, 111, 119kg respectively.

### 3.2.2 Rope wear/fatigue

The new ropes were then replaced with old ropes which are naturally worn after over 2 million times' running. Actually the old ropes are still in relatively good condition owing to fine lubrication. So rope wear/fatigue to certain extent is presented in this case. The comparison between old and new rope is given in Figure 3.6.

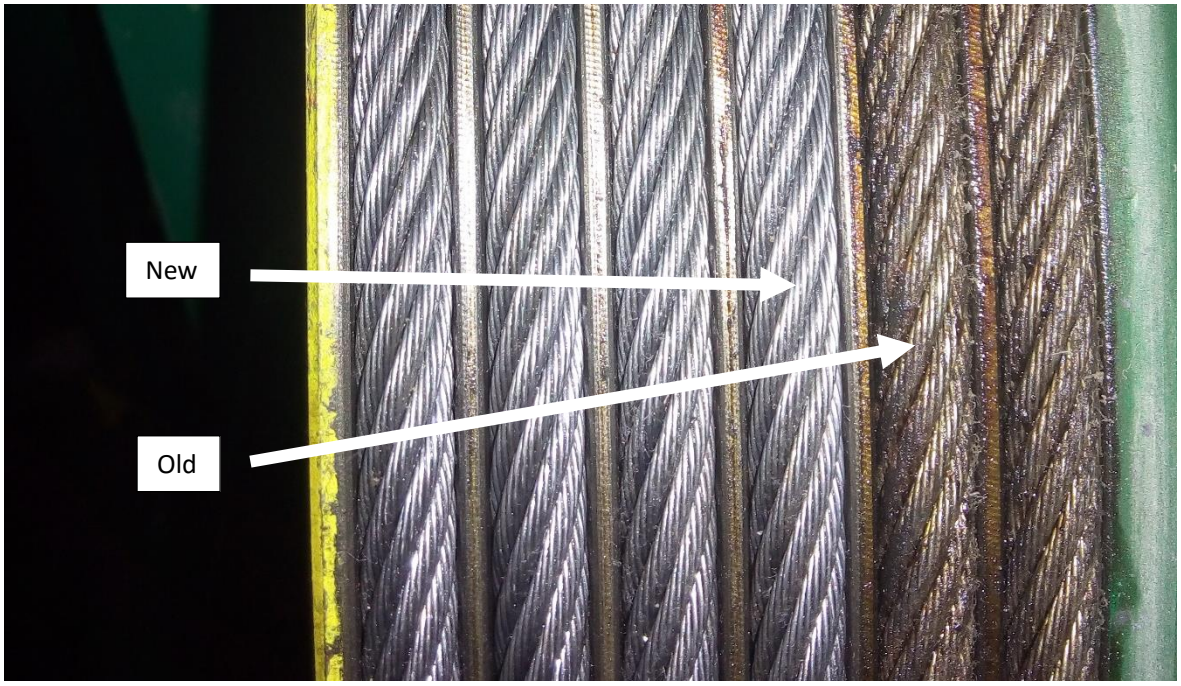



Fig 3.6 Rope wear/fatigue

### 3.2.3 Rope end loose

To mimic the condition of rope loose, we loosened 2 of all 6 ropes by adjusting their threaded connection, see Figure 3.7. The measured rope tension forces in this case are respectively 71, 93, 99, 72, 106, 129kg, lower than baseline case.

Fig 3.7 Rope end loose

### 3.2.4 Rope local fault

In order to study the vibration reflection of rope local defect, we introduced a rope local bump case by mounting a metal ring on one of the rope, as exhibited in Figure 3.8.

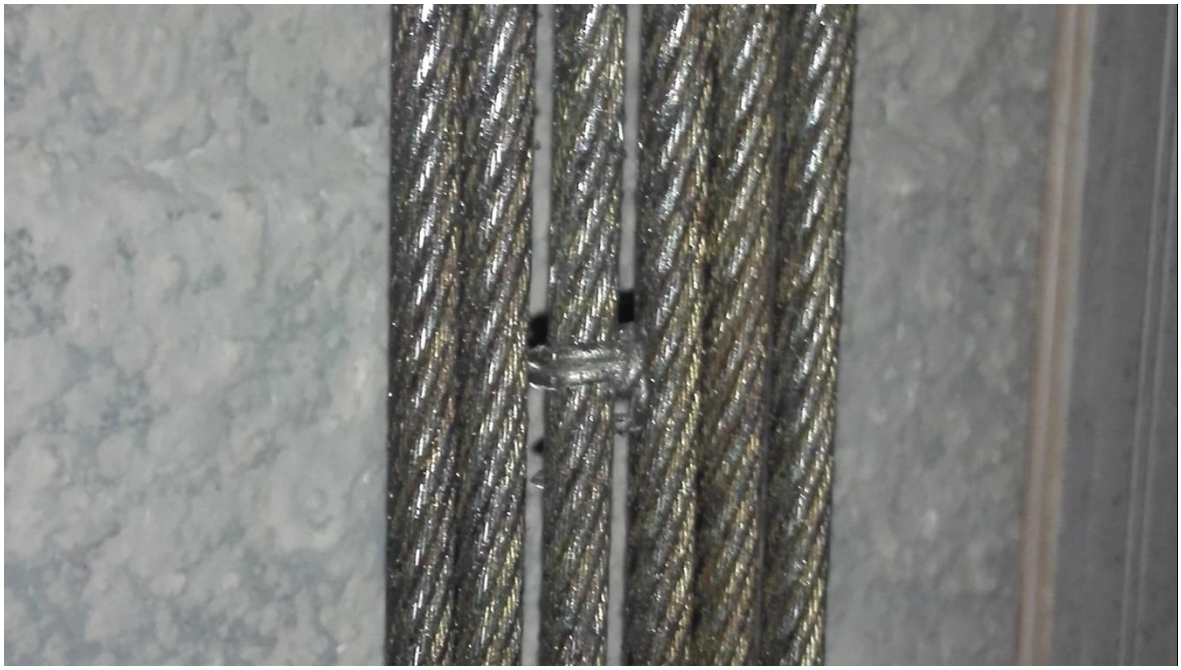


Fig 3.8 Rope local fault

## 3.3 Test scenario

### 3.3.1 Vertical modal test

Vertical modal test were conducted to obtain the system natural frequencies, which are determined by parameters like mass and stiffness and can work as an indicator in detecting rope loose fault. Since the parameters vary along with the travel, natural frequencies are actually nonstationary. To minimum the influence of varying parameters, we performed modal test on each floor at relatively fixed positions. Comparisons of modal test on same floor can theoretically provide meaningful information.

Initial excitation is the key to arise free natural vibration. For vertical modal test, since the common force hammer is small in size and unable to provide enough initial force, we came up with a new way. The car travel from 1-4 floor and braked suddenly at each floor height. By measuring elevator vibration with aforementioned sensors, we are able to analyze the frequency constitution of free vibration signal and thus identifying natural frequencies. The scenario of modal test is recorded in Table 3.1.

Table 3.1 Vertical modal test scenario

| No. | Location | Load (car frame) | Load (car) | Sampling time | Note | File |
|---|---|---|---|---|---|---|
| 1 | Floor 4 | 3*60kg | 0 | 10s | Brake | Xx01 |
| 2 | Floor 3 | | | | | |
| 3 | Floor 2 | | | | | |
| 4 | Floor 1 | | | | | |

### 3.3.2 Rope swing test

Rope swing frequency is a newly proposed frequency indicator for reflecting rope health condition. In order to obtain the rope swing frequency and evaluate its performance in diagnosing rope loose fault, rope swing test were conducted. Similarly, tests on both the top as well as the bottom of the building are included.

Since we aim to calculate the frequency value of swing frequency and pay little attention to its amplitude, here we introduced lateral stimulation to the rope by simply shaking the rope manually. The scenario of rope swing test is recorded in Table 3.2.

Table 3.2 Rope swing test scenario

| No. | Location | Load (car frame) | Load (car) | Sampling time | Note | File |
|---|---|---|---|---|---|---|
| 5 | Floor 1 | 3*60kg | 0 | 10s | Shake | Xx05 |
| 6 | Floor 4 | | | | | |

### 3.3.3 Car travel test

Car travel test is essential to mimic real-world running conditions and meanwhile validate the effectiveness of 1) using rope passing frequency to diagnose rope wear/fatigue fault, as well as 2) using rope swing frequency to detect rope local fault.

The building is merely 4-storey high, thus we let the car travel between 1$^{st}$ and 4$^{th}$ floor and ignore the shorter travel like 2$^{nd}$-3$^{rd}$ floor, etc. It should be emphasized that during travel test we manually control the car speed to 0.5m/s, hence a relatively stationary running condition is available.

With the same transducer distribution, we conducted car travel test under 2 different load conditions (light load and heavy load), and detailed scenario is given in Table 3.3.

Table 3.3 Car travel test scenario

| No. | Location | Load (car frame) | Load (car) | Sampling time | Note | File |
|---|---|---|---|---|---|---|
| 7 | Floor 4~floor 1 | 3*60kg | 0 | 60s | Smooth | Xx07 |
| 8 | Floor 1~floor 4 | | | | | |
| 9 | Floor 4~floor 1 | | | | | |
| 10 | Floor 1~floor 4 | | | | | |
| 11 | Floor 4~floor 1 | | | | | |
| 12 | Floor 1~floor 4 | | | | | |
| 13 | Floor 4~floor 1 | 3*60kg | 4*60kg | 60s | Smooth | Xx13 |
| 14 | Floor 1~floor 4 | | | | | |
| 15 | Floor 4~floor 1 | | | | | |
| 16 | Floor 1~floor 4 | | | | | |
| 17 | Floor 4~floor 1 | | | | | |
| 18 | Floor 1~floor 4 | | | | | |

# 4 APPLICATION 1: DETECTION OF ROPE WEAR

Hoist rope is one of the core component of elevator system, yet the highly stressed condition and potential mutual slip always fasten the surface wear and fatigue. The maximum allowable stress value decreases along with the deterioration of wear, which may bring in extra defect and lead to potential safety hazard. Thus, detection of rope wear, especially incipient coating or surface wear, is of significance.

## 4.1 Rope passing frequency

The rope passing frequency was discovered in studying the surface structure of rope. As illustrated in Figure 4.1, there are cyclic ridges and pits on the surface of rope. When the rope runs through a pulley, the contact stiffness varies at a frequency which can be cast as

$$f_{\mathrm{p}} = \frac{v}{l}, \tag{4.1}$$

where $f_{\mathrm{p}}$ stands for rope passing frequency, $v$ is the rope linear velocity and $l$ means the interval between adjacent ridges, i.e., lay length divide number of ropes.

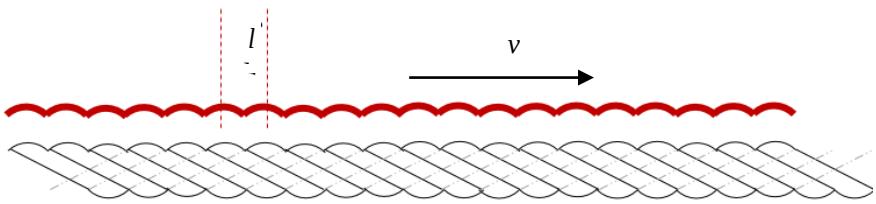



Fig 4.1 Rope surface structure

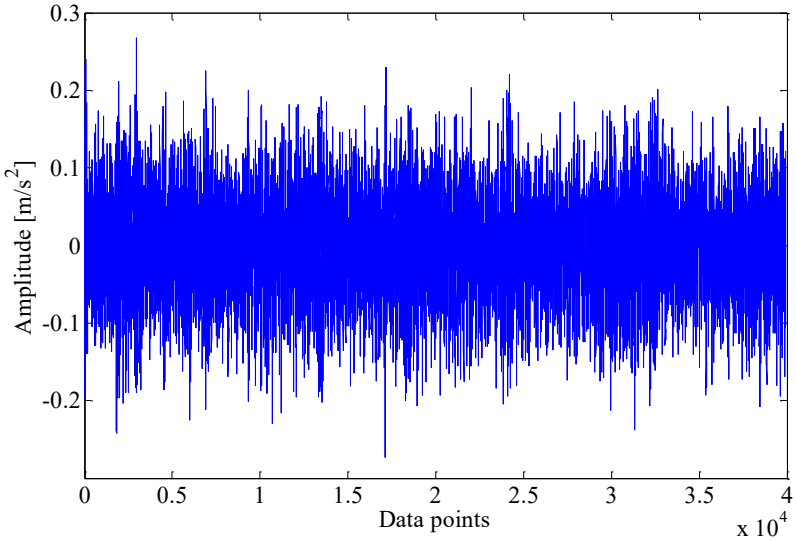
The value of rope passing frequency can be obtained once the elevator running speed and rope structure parameter are available. But the amplitude of this proposed indicator is determined by rope health condition as well as car load. In this test evaluation we loaded the pilot elevator with same mass and analyzed pulley vibration signals under both healthy and wear case to evaluate the performance of rope passing frequency.

## 4.2 Baseline

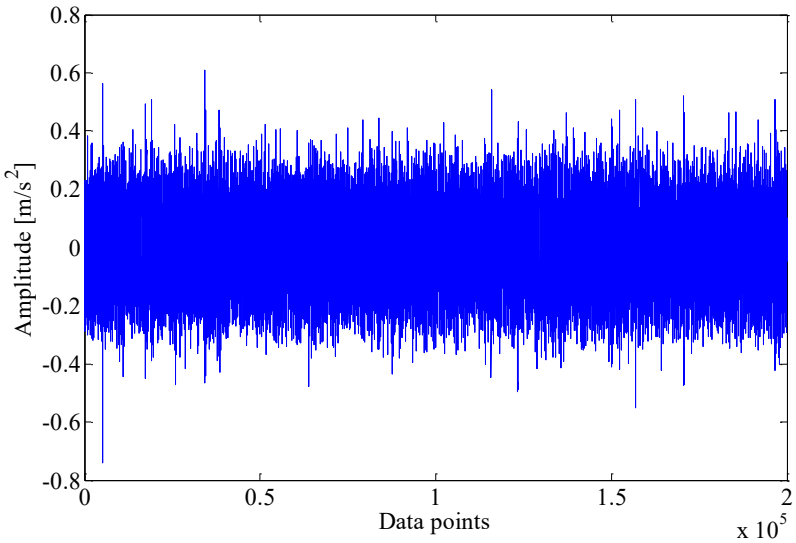
In baseline case 6 brand new hoist ropes were used, as shown in Figure 3.4. For an 8mm rope used in pilot elevator, its lay length is measured as 6mm. The running speed of rope is fixed around 1m/s. Therefore in pilot elevator test, the theoretical rope passing frequency is calculated as

$$f_{\mathrm{p}} = \frac{v}{l} = \frac{1}{6*10^{-3}}\,\mathrm{Hz} = 166.7\,\mathrm{Hz}\,. \tag{4.2}$$

Since hoist rope is in direct contact with car frame pulley as well as traction sheave, we analyze both the pulley radial directional vibration signal and motor vertical vibration signal during stationary car travel process. The waveform of baseline signals are plotted in Figure 4.2 (a) and (b).



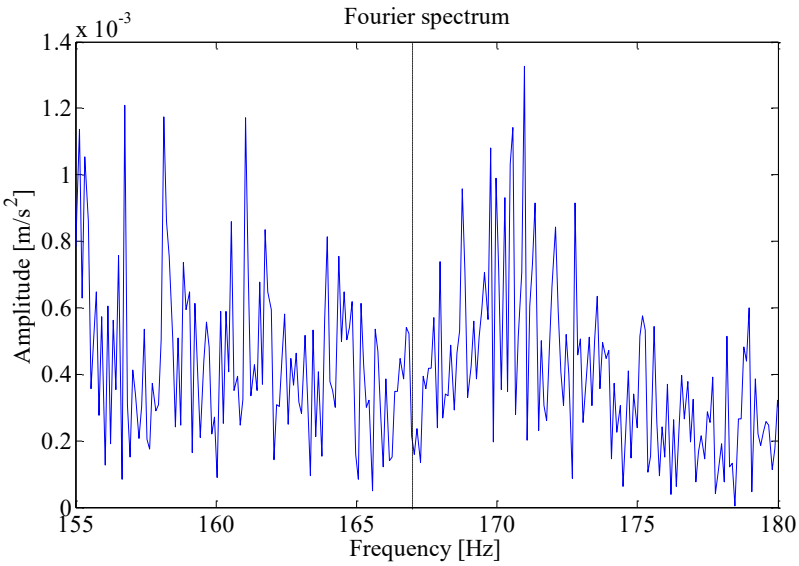

(a) Pulley vibration signal



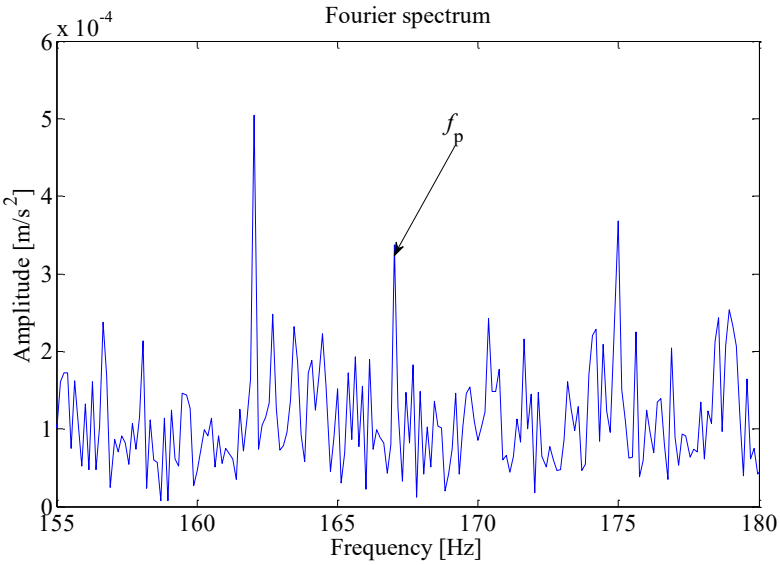

(b) Motor vibration signal

Fig 4.2 Waveform of baseline signal

We then generate the Fourier transform of these two signals and try to extract the rope passing frequency. A frequency range of 155Hz to 180Hz is presented which covers the rope passing frequency (167Hz), as exhibited in Figure 4.3.




(a) Pulley vibration signal



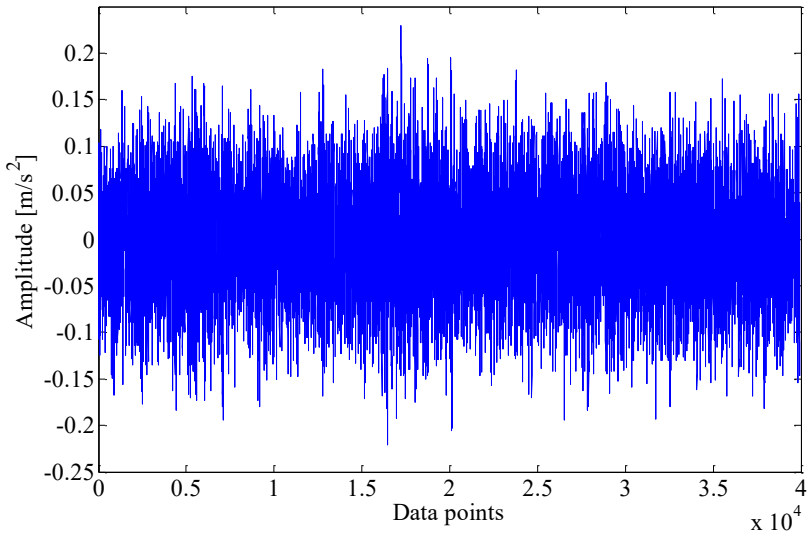

(b) Motor vibration signal

Fig 4.3 Fourier spectrum of baseline signal

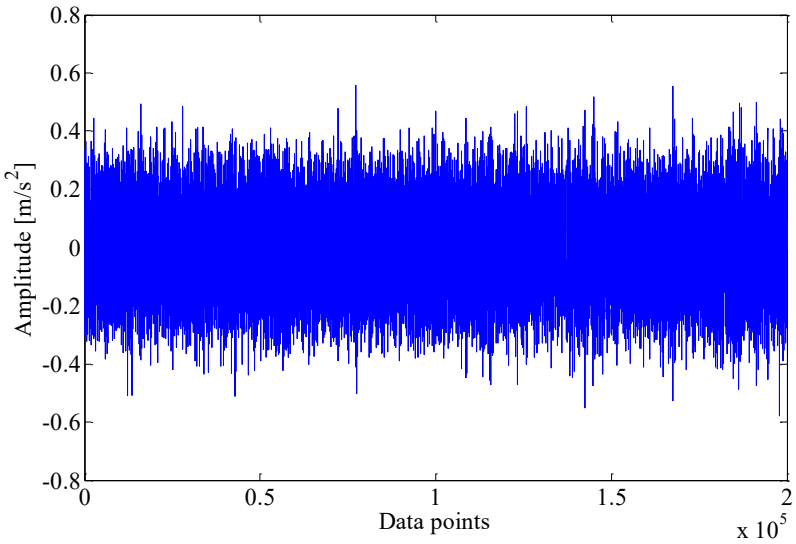
From Figure 4.3(a) we can barely identify the rope passing frequency, as labeled with dash line. Yet in the Fourier spectrum of motor vibration signal the 167Hz frequency peak is evident. Since the diameters of pulley and traction sheave are different, it's reasonable why the rope passing frequency is not equally prominent. Comparisons with corresponding signal Fourier spectrum in rope wear case should be done before conclusions are drawn.

## 4.3 Rope wear detection

In this case the old hoist ropes are mounted, as seen in Figure 3.5. Similarly, we conducted travel test under same stationary running speed and collected vibration signals from both pulley and traction sheave. Figure 4.4 (a) and (b) present the waveform of these two rope wear signals respectively.



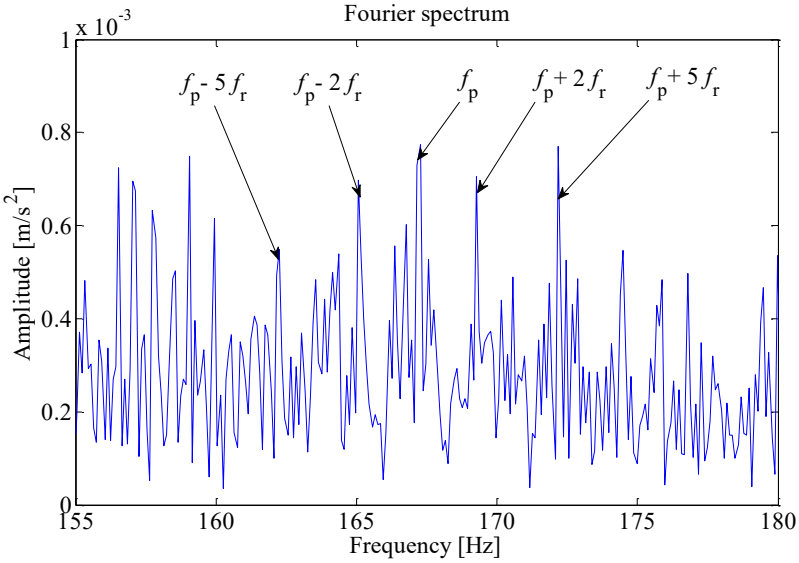

(a) Pulley vibration signal



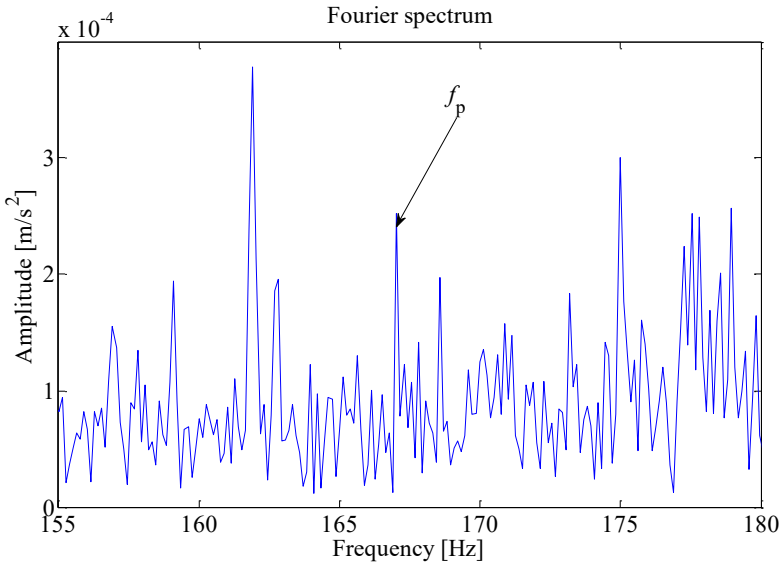

(b) Motor vibration signal

Fig 4.4 Waveform of rope wear signal

For comparison, the Fourier transform of these two rope wear signals are presented in Figure 4.5 (a) and (b). Again we focus on a frequency range of 155Hz to 180Hz and try to extract frequency components related to rope passing frequency.




(a) Pulley vibration signal




(b) Motor vibration signal

Fig 4.5 Fourier spectrum

In Figure 4.5(a) the rope passing frequency $f_p$ can be obviously pinpointed. Besides, clear sidebands are also extracted, which are further validated as the combination of rope passing frequency and pulley rotating frequency $f_p \pm 2f_r$ and $f_p \pm 5f_r$. It is an evident sign that periodical bump in hoist rope has stronger impact on the pulley vibration signal, indicating rope wear fault. For Fourier spectrum of motor vibration signal, the rope passing frequency is still visible, which is quite similar to baseline case. Since the diameter of traction sheave is bigger than that of pulley, this can be described as the assumption that in lower rope curvature, the rope passing frequency is more prominent.

## 4.4 Conclusion

On-site experiment on Jinan pilot elevator was carried out, in which both healthy rope case as well as rope wear case were included. According to the comparisons between spectra of both conditions, the rope passing frequency and its modulation with pulley rotating frequency are identified in wear case while not visible in baseline case. Therefore rope passing frequency is sensitive to rope wear condition and can be used to detect rope wear, especially incipient wear.

It is also found that the rope passing frequency is more prominent in motor vibration signal, mostly because of traction sheave's higher diameter. Considering the difference of sensitivity to rope wear fault, pulley vibration signal and motor vibration signal, or signals perceived from different rope curvature condition, are suggested to be further studied in lab experimental conditions.

# 5 APPLICATION 2: DETECTION OF ROPE LOCAL FAULT

Rope local faults like local bump or local strand breakage are reasons for noises and unintended car shake, which seriously influence the riding comfort. In this section, we study the rope local fault diagnosis method from the perspective of vibration. The stimulation from contact with local fault will result in rope swing, whose frequency is determined by the rope length and tension force. Therefore we propose to detect rope local fault by using rope swing frequency indicator.

## 5.1 Rope swing frequency

The rope swinging frequency can also be considered as rope radial natural frequency. As illustrated in Figure 5.1, when a local rope fault exist, rope swinging is excited. Since the rope swing natural frequency is fixed, we can diagnose the rope local fault by simply monitoring the energy of rope swinging frequency. The calculation of rope swinging frequency is derived as

$$f_s = \sqrt{\frac{Tn^2}{4ml^2}}, \tag{5.1}$$

where $f_s$ stands for rope swinging frequency, $T$ is the rope end tension, $n$ stands for the order of natural frequency, $m$ means the mass per unit length and $l$ is the corresponding rope length. It is also worth noticing that, since the vibration wave is generated from pulley and spread in both directions independently, two rope swing frequency components are supposed to be discovered in case of rope local fault.

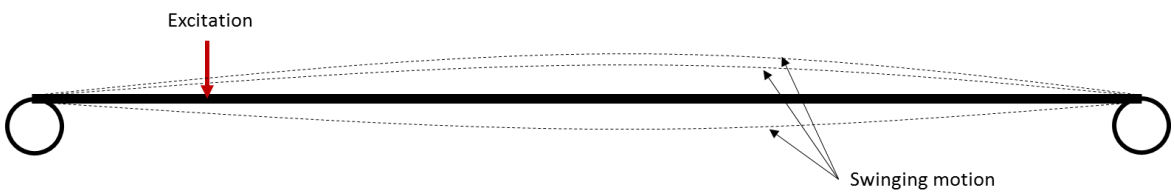



Fig 5.1 Rope surface structure

## 5.2 Baseline

In baseline case hoist ropes without obvious defects were used, hence no prominent rope swing frequency is supposed to be identified. Theoretically the accurate rope tension force, mass and rope length are demanded for rope swing frequency calculation, yet these mechanical parameters are not always available. Here we used experimental test method to estimate the rope swing frequency from rope's modal test result. The detail of rope swing test is already given in Section 3.3.2. After collecting the car frame vibration signal from swing test on 4th floor, we generate its time frequency representation (TFR) in Figure 5.2. The rope swing frequency is evident and validated as around 10Hz.

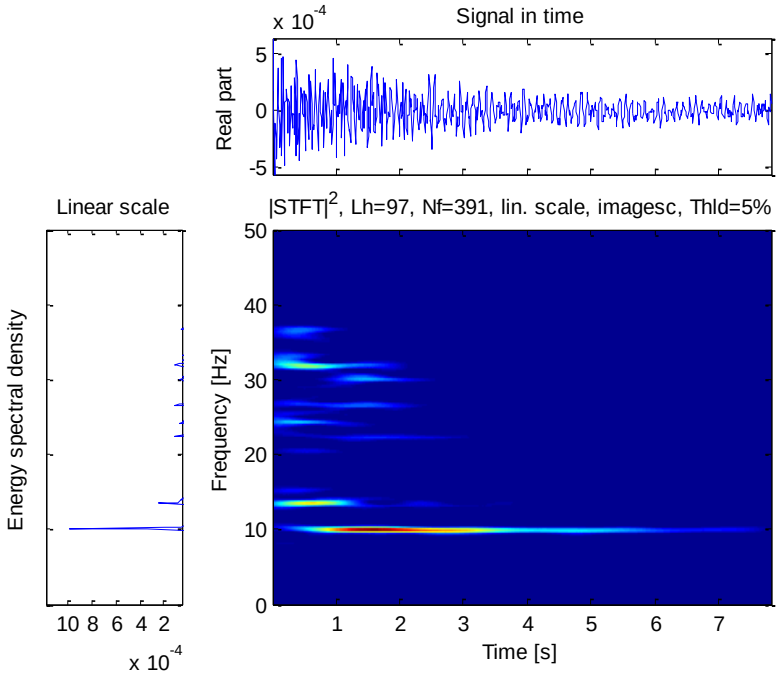
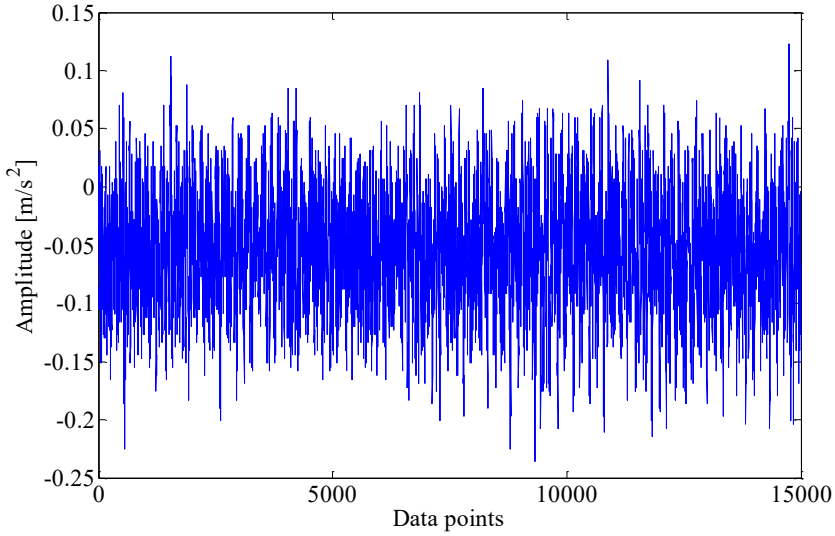
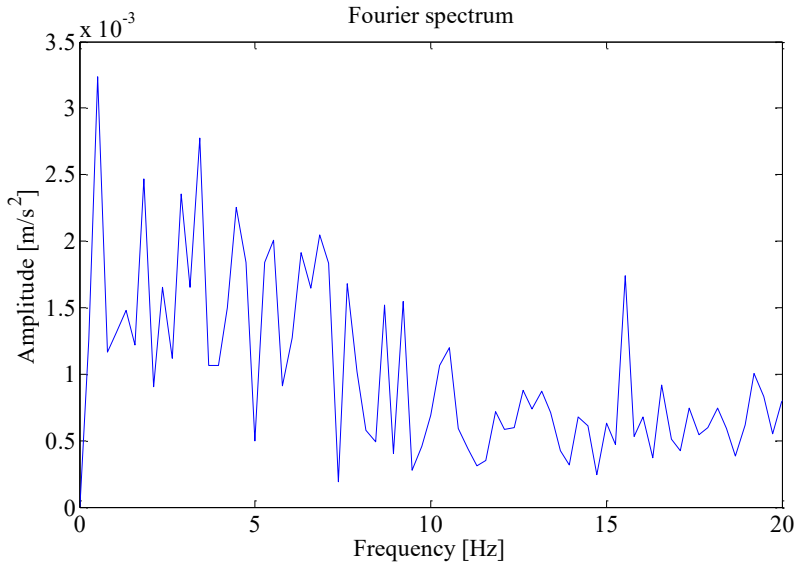



Fig 5.2 TFR of car frame vibration signal in rope swing test

Since rope local defects are always weak and it is hard to detect corresponding swing from car frame, we choose to monitor the vibration of pulleys which directly contact with ropes. During the car travel test from 4th floor to 1st floor, we collected vibration signals from pulley in radial direction, and the waveform & Fourier spectrum of collected signal within selected time period are plotted in Figure 5.3 (a) and (b).




(a) Waveform



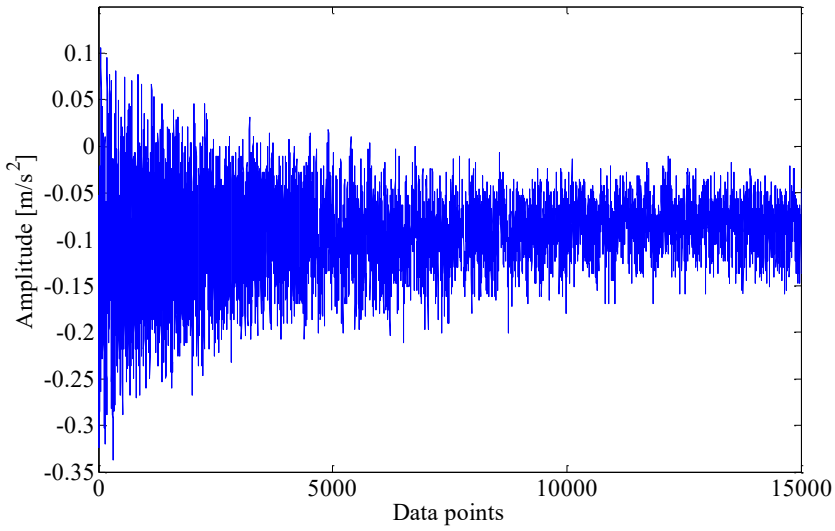

(b) Fourier spectrum

Fig 5.3 Pulley vibration signal in baseline case

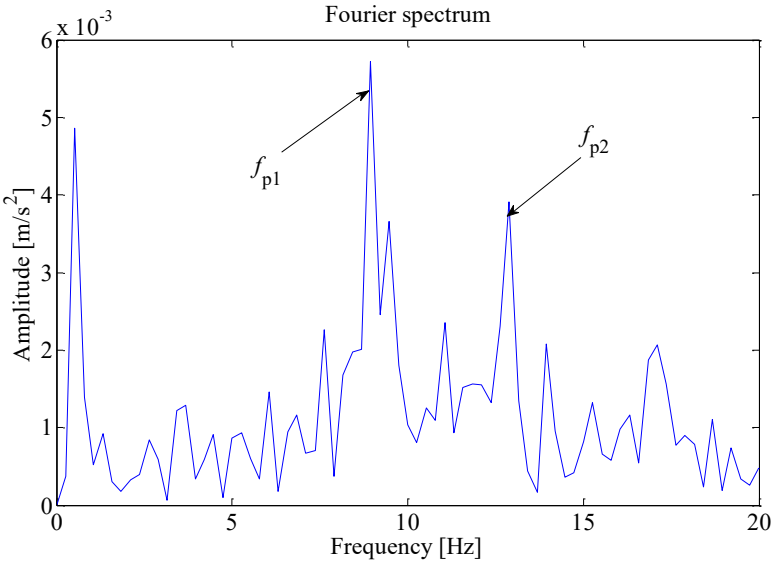
Since the rope swing frequency is around 10Hz, we focus on a frequency range of 0-20Hz to study the property of rope swing frequency in baseline case. In Figure 5.3(b), no clear frequency peaks are identified. It indicates that the hoist rope is relatively steady in lateral direction and no severe fault exist.

## 5.3 Rope local fault detection

In rope local fault condition we added a metal ring on one of the hoist ropes at the height of around 3rd floor, which can be found in Figure 3.7. Again car travel test from 4th floor to 1st floor was conducted, and the waveform as well as Fourier spectrum of car frame vibration signal after the impact resulted from local fault is exhibited in Figure 5.4.




(a) Waveform




(b) Fourier spectrum

Fig 5.4 Pulley vibration signal in rope local fault case

A frequency range of 0-20Hz is also focused on since the rope swing frequency is around 10Hz. Different from baseline case, in Fourier spectrum of pulley vibration signal under rope local fault condition, two frequency peaks are prominent. Since two parts of the rope swing at different frequencies (due to different length), these two frequency components around 10Hz are validated to be the rope swing frequencies. Therefore, the

rope local fault is successfully diagnosed. The position of the local fault can also be achieved by recording the time instant of the vibration impact.

## 5.4 Conclusion

Rope local fault is quite common under poor lubrication condition or when runs into accidental damage. How to effectively detect rope local fault is a challenging issue in ensuring the elevator riding comfort. On-site experiment on Jinan pilot elevator was conducted to demonstrate the effectiveness of rope swing frequency indicator in detecting rope local fault. Rope swing test was firstly carried out to estimate the numerical range of rope swing frequency from car frame vibration signal, after which car travel test was performed and the pulley vibration signal is collected.

In the Fourier spectrum of pulley vibration signal under rope local fault case, two rope swing frequency components were clearly identified, corresponding to the lateral vibration of two parts of the hoist rope. From the comparison of Fourier spectrum between baseline case and faulty case, the rope local fault can be successfully detected.

# 6 APPLICATION 3: DETECTION OF ROPE LOOSE

Hoist ropes are connected with fixed springs at both ends, which are also used to adjust the tension force. Traditionally a Henning device is required for each of the rope to measure corresponding tension force and guide the maintenance technician to check the balance of load distribution. Actually the rope loose has direct influence on the vibration properties of rope itself as well as the whole system. In this section both the rope swing frequency and system natural frequency are used for rope loose detection.

## 6.1 System natural frequency & rope swing frequency

Many of reported elevator accidents were due to rope loose or even derail. This category of rope connection fault has direct influence on system mechanical structure and result in the change of stiffness and other parameters. Hence, we study the characteristics of elevator system and propose the natural frequency as feature frequency indicator.

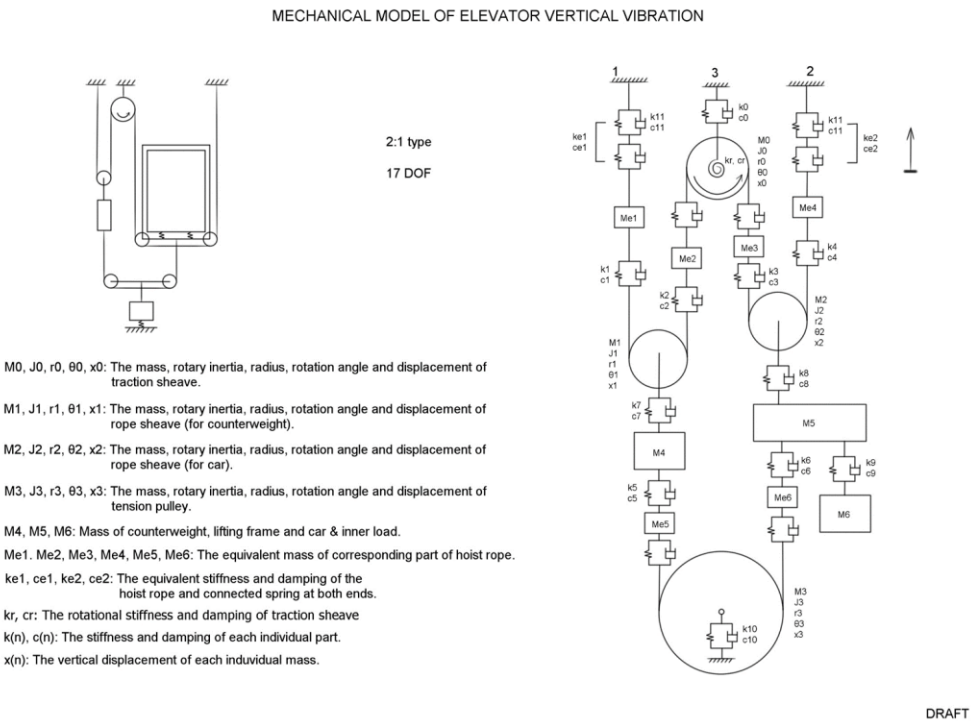



Fig 6.1 Vertical vibration model

According to the formerly proposed 17 DOF vertical vibration model as shown in

Figure 6.1, 17 natural frequencies theoretically compose of the system natural vibration signal. In case of rope loose, the stiffness of corresponding connection decreases and we have already demonstrated that system natural frequencies will correspondingly drop in proportional to the fault severity. Based on these rope loose mechanisms, we firstly propose the value of system natural frequency as indicator for rope loose diagnosis.

Another proposal is to exploit the aforementioned rope swing frequency (Section 5.1) in detecting rope loose fault. Since the swing frequency is directly determined by rope tension force, it is quite sensitive to rope loose. Besides, the vertical system natural frequency actually varies in small scale, and sometimes may bring in errors under poor frequency resolution conditions. The rope swing frequency is essentially lateral natural vibration, which is more sensitive to tension variation and thus facilitate the fault feature extraction.

## 6.2 Baseline

In baseline case 6 hoist ropes with relatively balanced load distribution were set. Both vertical modal tests and rope swing tests were conducted on each floor, as introduced in Section 3.3.

Firstly time-frequency analysis is performed on car frame vertical vibration signal in modal test to obtain the system natural frequency. The TFR of baseline car frame vertical vibration signal on 4$^{th}$ floor modal test is generated in Figure 6.2.

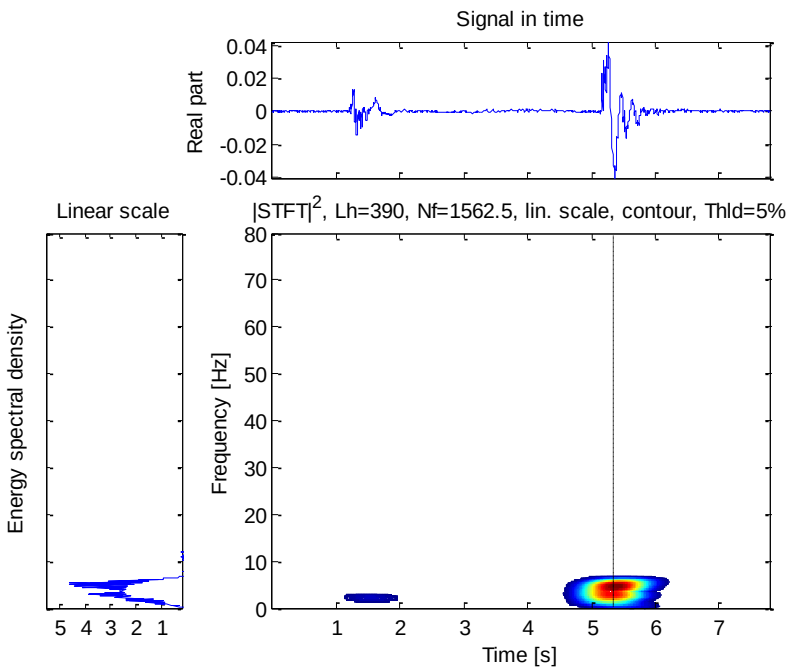



Fig 6.2 TFR of baseline car frame vertical vibration signal on 4$^{th}$ floor modal test

In order to make the TFR more readable and avoid the deviation of frequency value

reading, we further extract one column of the time-frequency matrix which covers the maximum amplitude, as labeled with dash line in Figure 6.2. The obtained amplitude vector is then plotted in a frequency spectrum as given in Figure 6.3.

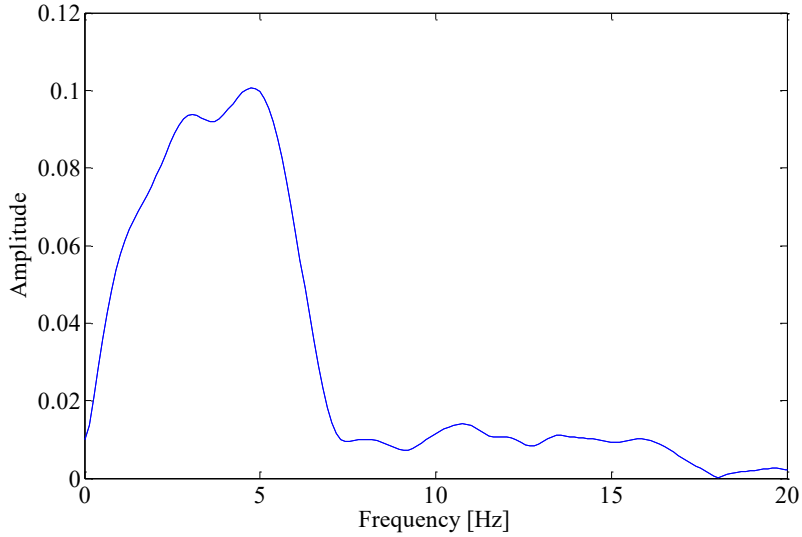



Fig 6.3 Spectrum of baseline car frame vertical vibration signal on 4th floor modal test

Similarly spectrum of baseline car frame vertical vibration signal on other floors are generated for comparison, as plotted in Figure 6.4 using different colors. It is evident that the prominent natural frequency varies along with the car height, and the value is larger on higher floors.

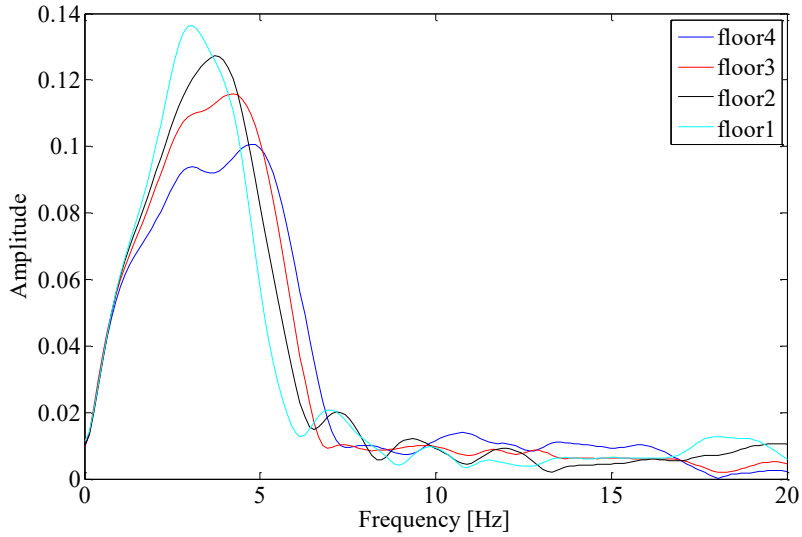
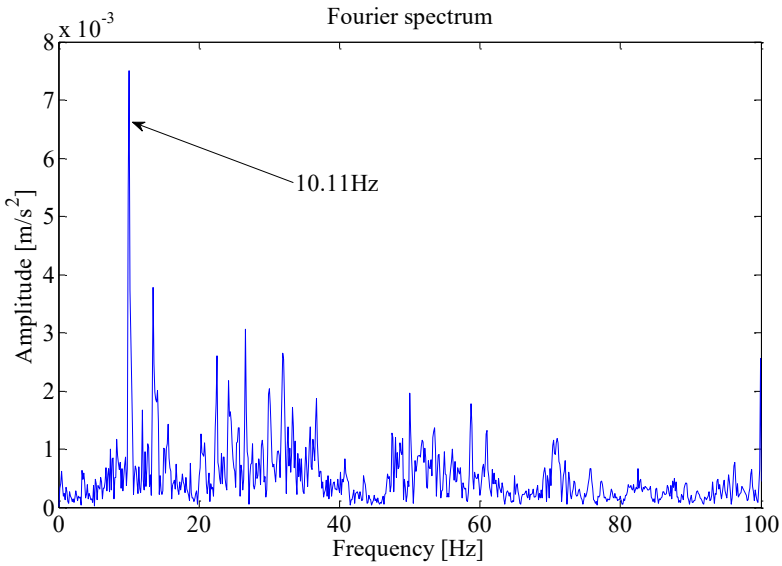


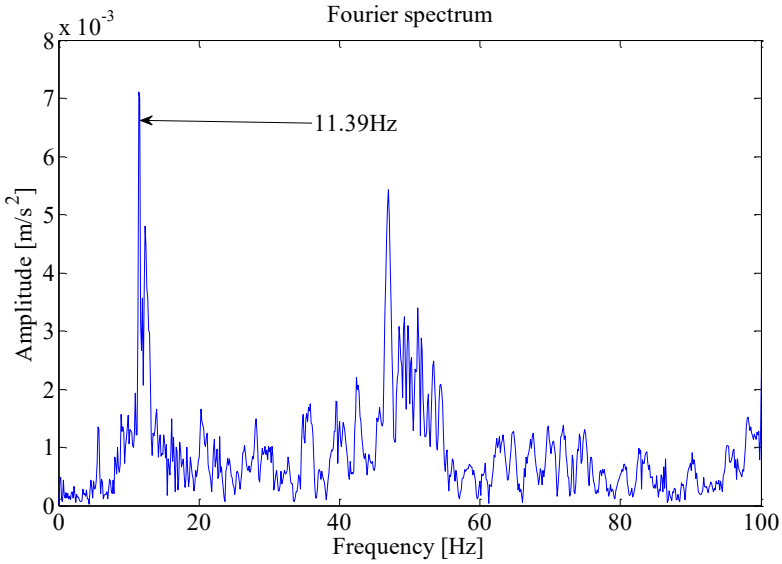

Fig 6.4 Spectrum of baseline car frame vertical vibration signal on modal test

In addition to the system natural frequency, we try to obtain the rope swing frequency from car frame swing test vibration signals. Since the rope swing frequency is constant on fixed height, Fourier spectrum is utilized to achieve rope swing frequency from baseline car frame vibration signals on swing test.

The Fourier spectra of baseline car frame vertical vibration signal on 1st floor and 4th floor swing tests are presented in Figure 6.5(a) and (b). It is easy to pinpoint the swing frequency under both circumstances, as 10.11Hz on 1st floor and 11.39Hz on 4th floor. These results will be compared with that in faulty cases.




(a) 1st floor




(b) 4th floor

Fig 6.5 Spectrum of baseline car frame vertical vibration signal on swing test

## 6.3 Rope loose detection

In rope loose case we manually loosen one of the hoist rope and carried out same modal test as well as swing test as did in baseline case.

Again we achieve the system natural frequency on each floor from time-frequency analysis of car frame vertical vibration signal on modal test. As given in Figure 6.6, the varying trend of system natural frequency along with car height is still evident. However,

comparing the obtained system natural frequency in baseline and rope loose case (Figure 6.4 comparing with Figure 6.6), we find the corresponding natural frequency relatively lower in rope loose case than in baseline, but in limited scale. This is most probably due to the restriction of on-site conditions and relatively weak fault introduction. Therefore, the sensitivity of system natural frequency still requires solid demonstration.

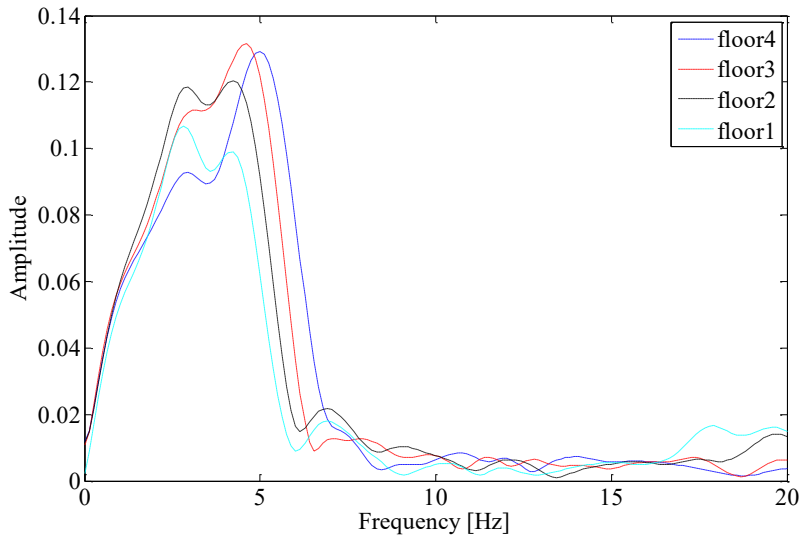
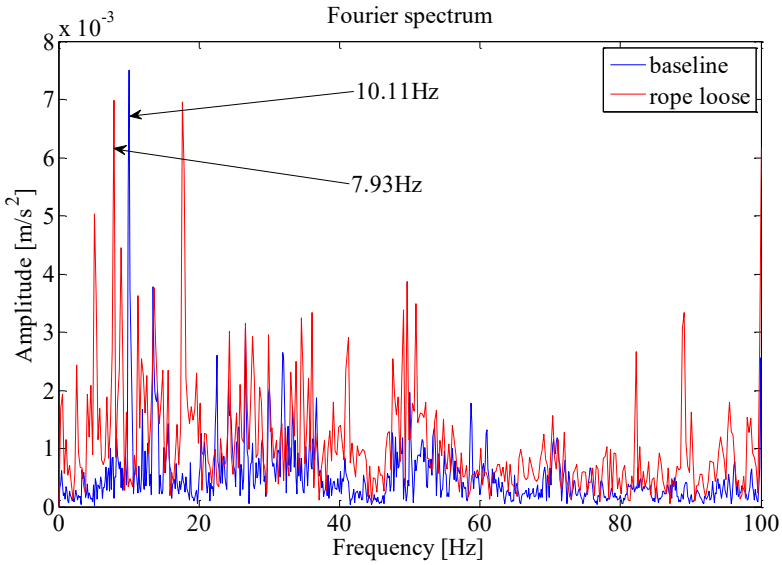
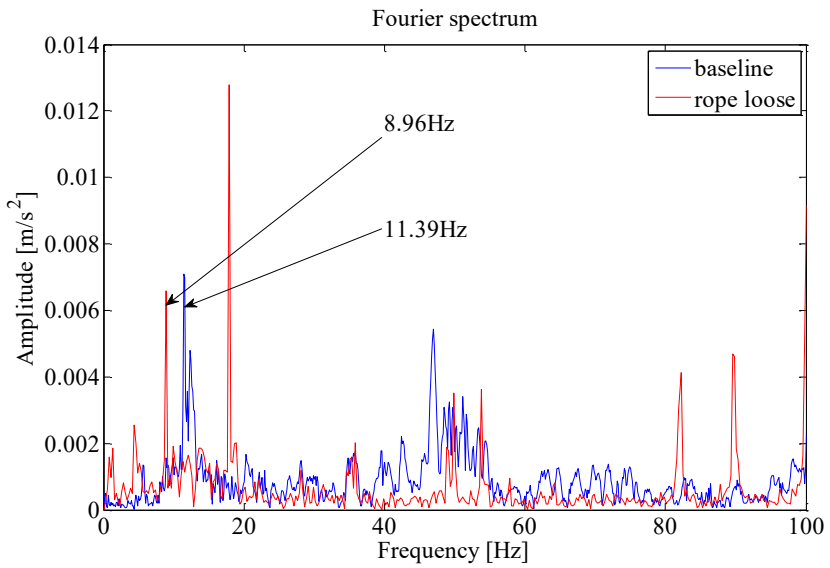



Fig 6.6 Spectrum of rope loose case car frame vertical vibration signal on modal test

Then Fourier spectra of rope loose case car frame vertical vibration signal on swing test are calculated. They are plotted in same figure with that in baseline case for comparison, as exhibited in Figure 6.7(a) and (b).




(a) 1st floor




(b) 4th floor

Fig 6.7 Spectrum of baseline car frame vertical vibration signal on swing test

As evident in Figure 6.7, the rope swing frequency in rope loose fault case is lower than that in baseline case, dropping from 10.11Hz to 7.93Hz on 1st floor and from 11.39Hz to 8.96Hz on 4th floor. Since the value of rope passing frequency is proportional to rope tension force, the decrease of rope passing frequency indicate rope loose fault. By detecting the value of rope passing frequency, we are able to diagnose potential rope loose fault without extra Henning system or complex experiments.

## 6.4 Conclusion

For rope loose fault diagnosis, two frequency indicators including system natural frequency and rope swing frequency are proposed. Both vertical modal test and rope swing test were conducted to evaluate their effectiveness and properties.

System natural frequency can be clearly pinpointed in time-frequency representation of car frame vibration signal in modal test. The value of natural frequency and its variance along with car vertical position is clearly presented in proposed spectrum derived from TFR. However, it decreases little when one rope loose is brought in, which may result from relatively weak fault severity. Further research of system natural frequency on its sensitiveness to rope loose fault severity are required before it can be applied in real world rope fault diagnosis.

Fortunately, rope swing frequency, which can be obtained from the Fourier spectrum of car frame vibration signal in rope swing test, is validated to be of effectiveness in detecting rope loose. By recording the rope swing frequency obtained from swinging of each rope, we are able to reveal the tension condition of hoist ropes without extra devices.

# 7 FAQS ON INSTRUMENT AND MEASUREMENT

A whole set of vibration measurement and analysis system composed of:

1) Vibration source (elevator/escalator)
2) Sensors (accelerometers, microphones, speed sensor, torque sensor, impact hammer, note that low noise cables are also important)
3) Data acquisition system
4) Analysis software and programme

In this section we introduce some basic rules for selecting appropriate sensors, data acquisition system and configuration parameters for vibration measurement of elevator, by answering some FAQs.

## 7.1 Sensors

**(1) Why we choose vibration-based measurement and analysis method for elevator monitoring and diagnosis?**

Vibration-based condition monitoring and fault diagnosis has long been applied in almost all industrial fields. It allows the maintainer to collect health information without shutting down the whole machine set. Besides, it is relatively flexible, sensitive to potential fault and does not involve complex devices.

**(2) What kind of accelerometers are suggested in elevator vibration measurement?**

Piezoelectric sensors are most extensively used accelerometers in industrial fields, owing to its wide frequency bandwidth, high sensitivity, robust reliability and simple structure. Two categories of piezoelectric sensors, including traditional ones and IEPE types, are both available. Traditional piezoelectric sensors have independent external conditioning circuits while IEPE ones integrate these circuits inside of the sensor. Obviously the IEPE sensors are much more convenient in on-site test. However, traditional piezoelectric sensors are still used today for a reason. In some extreme conditions like high temperature (over 150℃), circuit will easily melt and it is better to place all conditioning circuits elsewhere. Consequently, for elevator vibration test, IEPE piezoelectric accelerometers are suggested.

**(3) What parameters are most important in choosing an appropriate accelerometer?**

Usually there is a long list of parameters of each type of accelerometer. Except for basic demands like suitable measurement range and output voltage, to the best of our knowledge, two main characteristics are most significant in ensuring the correctness of perceived signals.

Sensitivity, or linearity of accelerometers is an essential parameter. It reflects how well the output voltage is linearly proportional to the input. Theoretically every climb in acceleration should meet same amount of increase in output voltage, yet this constant proportion is not always available even in extreme high/low frequency ranges. Generally the sensitivity is defined as the maximum deviation between actual measurements and fitted straight line, divide the full scale output. Therefore it is suggested to use accelerometers with relatively low sensitivity value. An example of sensitivity curve is shown in Figure 7.1.

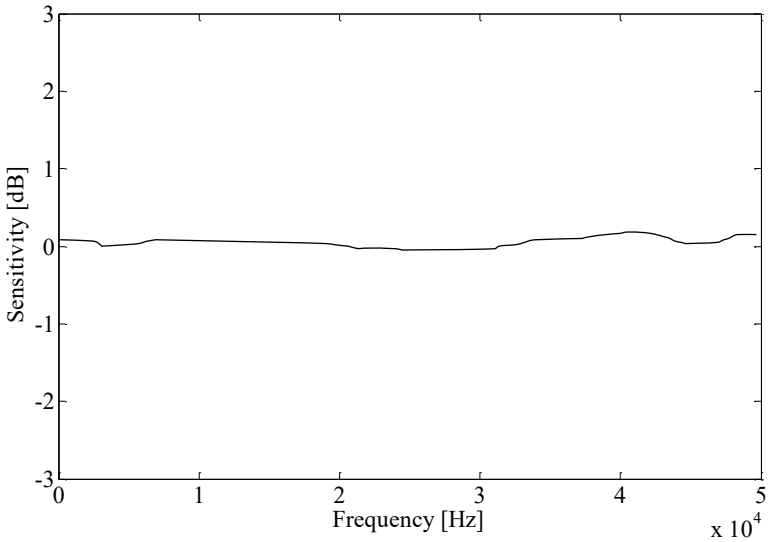



Fig 7.1 Sensitivity plot

Another key parameter is the amplitude-frequency response. It shows the ratio of amplitude of output signal over the amplitude of input signal. Generally only the frequency range with relatively constant amplitude-frequency response can be exploited in measurement and analysis, while accelerometers with relatively wide exploitable frequency range is preferred in real application. An example of amplitude-frequency response curve is presented in Figure 7.2. The reason why accelerometers may not be able to measure vibration at too low frequency is because the circuit can hardly process the perceived current signal which varies too slowly. And resonance phenomenon is inevitable since accelerometer itself can be considered as an vibration system.

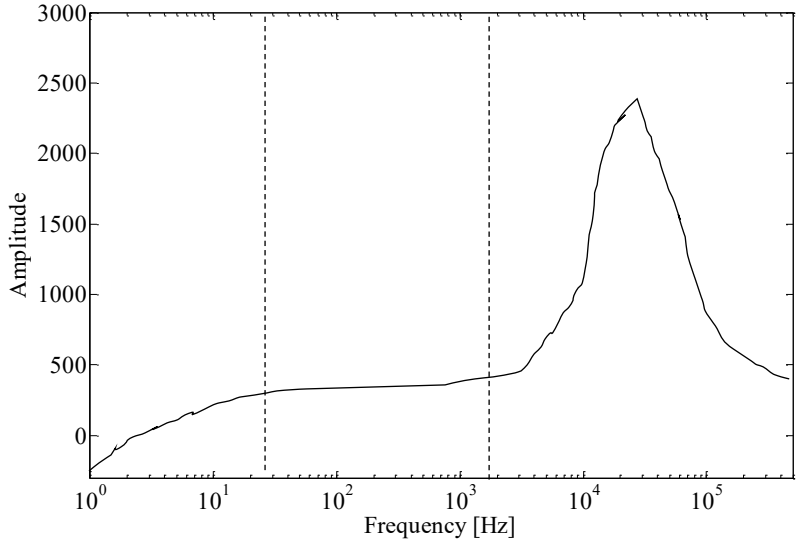



Fig 7.2 Amplitude-frequency response plot

**(4) What kind of sensors are also needed except for accelerometer? Are there recommended manufacturers/types?**

Accelerometers are essential for vibration measurement. For elevators, some important components like pulleys are not always within reach therefore wireless accelerometers are strongly recommended. Except for accelerometers, other sensors like microphones, tachometer and impact hammer are also useful to completely reflect the health information. Microphones are useful to collect the background noise and analyze its influence to passenger's comfort. Tachometer, which records the rotating frequency of traction motor, provides possibility of calculating the value of characteristic frequencies and car position. A impact hammer is necessary to perform modal test on components like motor or guide rail, or even the whole elevator system, which reveals the natural frequency properties.

Some recommended types are given as follows.

1) **Accelerometers (wired)**

- Considerations: **sensitivity**, **frequency range**, output voltage (match with DAQ system)
- Triaxial accelerometers may be convenient in test, but according to our experience individual signal axial one may have better performance.
- e.g. Model 623C01/ 393B12 by PCB Group, Inc.
- http://www.pcb.com/

2) **Accelerometers (wireless)**

- Considerations: sensitivity, frequency range, size, output voltage,
- e.g. BND-AX3D-10G by BeanAir, 牧坤 M67, JM5843 by 扬州晶明

3) **Microphones:**

- Considerations: sensitivity, frequency range, output voltage, size
- e.g. Model 378A06 by PCB Group, Inc.
- http://www.pcb.com/

4) **Tachometer:**

- Considerations: speed range, IEPE current excitation
- e.g. ICP® Laser Tachometer — LaserTach

5) **Impact hammer:**

- The size of the objective structure have to be taken into consideration
- e.g. PCB Impact hammer
- http://www.pcb.com/testmeasurement/impacthammers

## 7.2 Data acquisition system

**(1) What is data acquisition system? What is A/D conversion?**

In general, data acquisition system receives the analog voltage signal from sensors and convert it to recognizable digital signals for computer. Analog quantities indicate those continuously varying quantities, which covers almost all natural signals. However, all these analog signals have to be converted to digital ones which can be stored and analyzed by computers. This is the reason why we need a DAQ system in vibration measurement.

**(2) What are the main considerations in choosing an appropriate DAQ system?**

Surely the speed of signal conversion is important in evaluating the performance of DAQ system. However, no clear difference seems to exist between models at similar price. In choosing an appropriate DAQ system for measurements, one should refer to parameters like channel number, maximum sampling frequency and data resolution.

In lab experimental vibration tests of elevator, over 12 channels in total are recommended in order to collect enough data for comparisons and validations. Yet the measuring points are barely concentrated and more than one DAQ systems with less

channels for signal system should work. For example, in former tests in Beijing, Jinan and Kunshan, 3 sets of DAQ systems with 4 channels on each one were actually used.

Maximum sampling frequency is another parameter worth consideration. Since the frequency range in analysis is 0 to half the sampling frequency, we should set the sampling frequency higher than twice the maximum value of characteristic frequencies. Besides, the Shannon sampling frequency also tells us to collect data at sampling frequency no lower than twice the maximum value of characteristic frequencies. In real application we wish to measure vibration at a relatively high sampling frequency, e.g., 20000Hz, to obtain as much potential information as possible. The collected signal can be resampled and analyzed by signal processing software afterwards.

The data resolution means how many bits of binary numbers are used to store a single data. Since the measurement range is fixed, the data resolution is determined by the number of digits. Usually a 24-bit resolution is provided.

**(3) Are there recommended manufacturers/types?**

1) LMS SCADAS (Made by Siemens)

- Advantage: Be of high quality, many options
- Disadvantage: Relatively expensive
- Either SCADAS Mobile or SCADAS Recorder is fine;
- Under tough working conditions or deal with complex systems, SCADAS Recorder which can be wirelessly connected to PC can ensure the safety of testers.
- At least 8 channels are required in one DAQ system;
- At least 2 DAQ systems may be needed for overall test (we used 2 in former measurement)
- http://www.plm.automation.siemens.com/zh_cn/products/lms/testing/scadas/index.shtml

2) NI DAQ (Made by National Instruments Corporation)

- Advantage: Light and convenient, relatively cheap
- Disadvantage: resampling frequency bias, limited channel, limited module
- Several test module can be integrated in a chassis;
- Wireless DAQ chassis available;
- Two or more DAQ systems with over 16 channels in total are demanded;
- http://sine.ni.com/np/app/main/p/bot/no/ap/daq/lang/zhs/pg/1/sn/n17:daq,n24:cDAQ,n36:11/
- http://www.ni.com/data-acquisition/compactdaq/zhs/

**(4) Are there recommended measurement & analysis software?**

DAQ systems also need to be connected to matching software for data storage. Generally manufacturers of DAQ system will develop their own bundled software. For example LMS SCADAS made by Siemens is suggested to run with LMS Test.Lab software and NI DAQ can be controlled by Compact DAQ software.

There is no doubt that bundled software can be connected with corresponding DAQ system seamlessly. However, most of these software can only run some basic signal analysis method like waveform display, Fourier spectrum analysis and calculating basic time domain indicators, and much of the parameters in these analysis cannot be modified manually.

Since a series of time domain analysis (waveform, correlation analysis and different indicators), spectrum analysis as well as time-frequency analysis are involved in the process of R&D, Matlab is recommended as the main analysis software. Modeling and various numerical simulation can also be realized on Matlab platform, and corresponding programme based can be conveniently modified and transplanted.

# 8 INTRODUCTION OF TIME-FREQUENCY ANALYSIS IN ELEVATOR MONITORING & DIAGNOSIS

Elevator system is essentially a complex multiple degree of freedom nonstationary system. Its nonstationarity mainly comes from 3 aspects.

1) The car running speed is time-variant to provide smooth start and stop process. Therefore all characteristic frequencies related to rotary parts like motor and pulleys, are time-variant;
2) As the car travels, lengths of each section of hoist ropes varies simultaneously. Since the mass and stiffness parameters are time-variant, system natural properties are nonstationary;
3) In real applications the number of passengers are not fixed, hence generally the elevator system runs under nonstationary load conditions.

Due to the changing characteristic frequencies and perturbations, conventional Fourier spectrum analysis fail to thoroughly reveal the accurate time–frequency structure, resulting in indistinct wide band feature. Therefore, time-frequency analysis (TFA) is suggested here for analyzing nonstationary elevator signals.

## 8.1 Linear time-frequency representations

### 8.1.1 Short-time Fourier transform

Short time Fourier transform adds a time variable to the traditional Fourier spectrum, thus allowing it to investigate the time-varying nature of a signal. With this method, a short time window moves along time to slice a signal. In a short duration, the segmented signal usually does not change too much and hence can be assumed to be stationary. By applying the Fourier transform, the local Fourier spectrum of each segment around the time location of the short time window can be obtained and the time variation of the signal can be revealed according to the local spectrum.

The short time Fourier transform is obtained as follows. For any signal $x(t)$, suppose $w(\tau - t)$ is a window function centered at time $t$ (where $\tau$ is a time variable). Then the observed or segmented signal through this window is $x(\tau)w(\tau - t)$. Sliding the window

through the time span of interest and applying the Fourier transform to each segment leads to the short time Fourier transform

$$\mathrm{STFT}_x(t, f) = \int_{-\infty}^{+\infty} x(\tau) w(\tau - t) \exp(-j2\pi f \tau) d\tau \ . \tag{8.1}$$

Once the window function and its length are chosen, the time–frequency resolution of the short time Fourier transform is fixed. In applications, if we wish to closely localize the time instant of higher frequency components, a shorter time window should be used. If we would like to pinpoint the frequency location of lower frequency components, a longer time window is preferred. Due to the Heisenberg uncertainty principle, the finest time location and the best frequency resolution cannot be reached at the same time. Hence, the short time Fourier transform lacks adaptability. As a result, it is only suitable to analyze quasi-stationary signals that are stationary at the scale of the short time window, but is not appropriate to be used to analyze highly transient phenomena in signals, like impulses.

### 8.1.2 Wavelet transform

The wavelet transform employs wavelets, instead of sinusoidal functions, as the basis. It adds a scale variable in addition to the time variable in the inner product transform. Hence, it is effective for time–frequency localization, and is suited to transient signal analysis.

For any energy limited signal $x(t) \in L^2(R)$, the wavelet transform can be defined as

$$\mathrm{WT}_x(t, a) = \frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} x(\tau) \psi(\frac{\tau - t}{a}) d\tau \ . \tag{8.2}$$

where wavelet $\psi[(\tau - t)/a]$ is derived by dilating and translating the wavelet basis $\psi(t)$, $a$ is the scale parameter ($a > 0$), $t$ is the time shift, and $1/\sqrt{a}$ is a normalization factor to maintain energy conservation.

The variable time localization and frequency resolution nature enables the wavelet transform to have a zooming and adaptive windowing capability suited for nonstationary signal analysis. Wavelet transform iteratively decomposes the approximation signals of lower frequency, but does not further work on the detail signals of higher frequencies. For higher frequency components, wavelet transform has a better time localization but a lower frequency resolution. For lower frequency components, the frequency resolution is higher whereas the time localization is worse. Fortunately, the wavelet packet transforms can be employed to improve the frequency resolution for higher frequency components. The wavelet packet transform further decomposes the detail signals of higher frequency iteratively, i.e., decomposes a signal into multiple equi-bands, leading to better resolutions in both the time and frequency dimensions. Wavelet transform is essentially a time–scale analysis method and thus is effective in analyzing self-similar or fractal signals. To date,

various types of wavelet basis have been proposed. However, the question of how to choose a suitable one among them to match the signal structure remains an open issue. There is still a lack of generally accepted effective methods to address this issue.

## 8.2 Bilinear Time–Frequency Distribution

Bilinear time–frequency distribution represents the signal energy distribution in the joint time–frequency domain. The Wigner-Ville distribution is the basis of almost all the bilinear time–frequency distributions. It has the highest time–frequency resolution. However, for multi-component signals, it suffers from the inevitable cross-term interferences, thus is not suitable for many real applications. In order to maintain a higher time–frequency resolution, to obtain a non-negative distribution, and to suppress the cross-term interferences in the Wigner-Ville distribution, improved versions such as Cohen class distributions, affine class distributions, adaptive optimal kernel method and re-assignment method have been proposed. Here only the basic theory of Wigner-Ville distribution are introduced.

For any signal $x(t)$, the Wigner-Ville distribution is defined as

$$\begin{aligned}\mathrm{WVD}_x(t,f) &= \int_{-\infty}^{+\infty} x(t+\frac{\tau}{2})x^*(t-\frac{\tau}{2})\exp(-j2\pi f\tau)d\tau \\ &= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} A_x(\tau,\upsilon)\exp[-j2\pi(\upsilon t+f\tau)]d\upsilon d\tau\end{aligned} . \tag{8.3}$$

where $A_x(\tau,\upsilon)$ is the ambiguity function given by

$$A_x(\tau,\upsilon) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} x(t+\frac{\tau}{2})x^*(t-\frac{\tau}{2})\exp(-j2\pi\upsilon t)dt . \tag{8.4}$$

As the Wigner-Ville distribution is not linear, the Wigner-Ville distribution of a sum of multiple signal components is not equal to the sum of the Wigner-Ville distributions of these signal components. For example, the Wigner-Ville distribution of signal $x(t) = x_1(t) + x_2(t)$ is

$$\mathrm{WVD}_x(t,f) = \mathrm{WVD}_{x_1}(t,f) + \mathrm{WVD}_{x_2}(t,f) + 2\,\mathrm{Re}[\mathrm{WVD}_{x_1x_2}(t,f)] . \tag{8.5}$$

Where

$$\mathrm{WVD}_{x_1x_2}(t,f) = \int_{-\infty}^{+\infty} x_1(t+\frac{\tau}{2})x_2^*(t-\frac{\tau}{2})\exp(-j2\pi f\tau)d\tau . \tag{8.6}$$

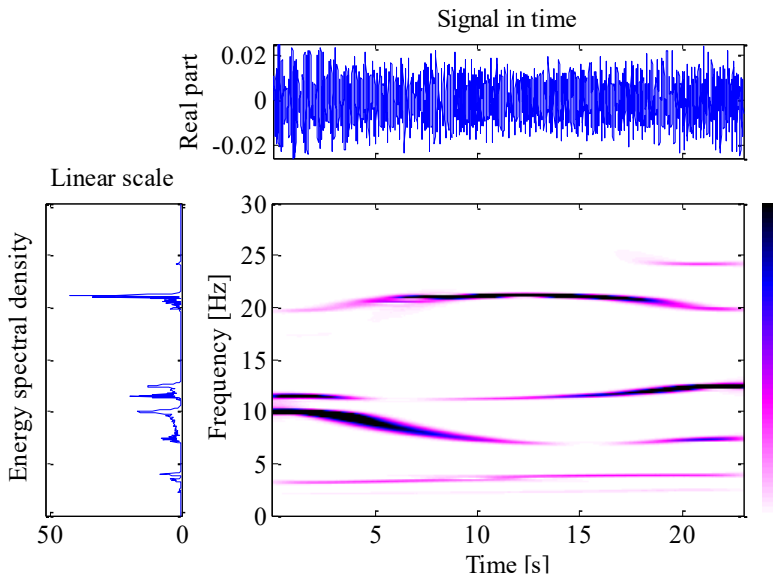
In Equation 8.5, the first two are auto-terms, and the third one is the cross-term. The cross-term results from the interference between time and frequency due to the nonlinear quadratic transform. It is real, and exists between the auto-terms. Moreover, the cross-term is oscillating and its amplitude is twice that of auto-terms. Each pair of auto-terms results

in one cross-term. If a signal has $n$ constituent components, there will be $C_n^2 = n(n-1)/2$ cross-terms. For multi-component signals in real applications, cross-terms produce a pseudo distribution and may overlap with the auto-terms on the time–frequency plane. In this case, the interference is even more complicated leading to blurred time–frequency features and making it more difficult to interpret the physical meaning of the Wigner-Ville distribution.

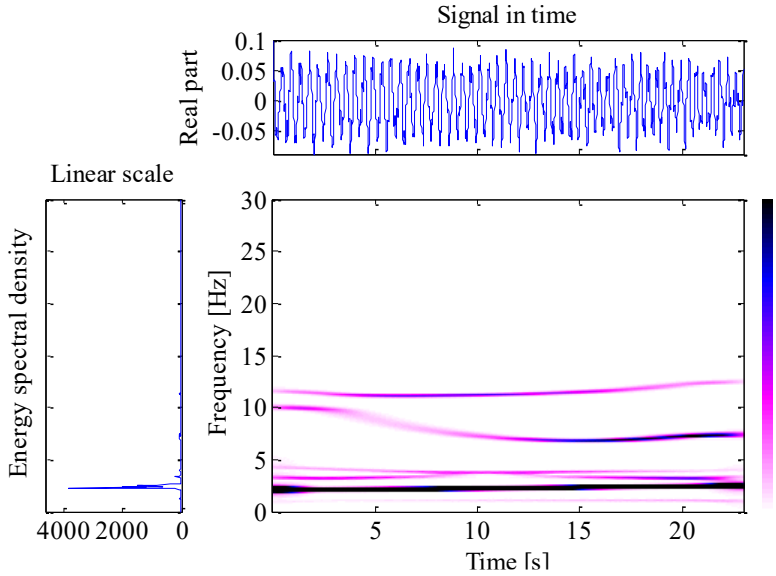
The Wigner-Ville distribution does not involve any window function and thus is free from the interference between time localization and frequency resolution. The product of time interval and frequency bandwidth reaches the lower bound of the Heisenberg uncertainty principle. Consequently it has the highest time–frequency resolution among all the time–frequency distributions. However, for multi-component signals in mechanical systems, its applications are inevitably hindered by the cross-term interferences. Even so, the fine time–frequency resolution of the Wigner-Ville distribution is still very attractive. Furthermore, as the cross-terms also contain information about signal structures, the question of how to effectively exploit the properties of cross-terms to reveal useful information could be an interesting topic.

## 8.3 Applications

Time-frequency analysis was first applied to analyze the varying trend of system natural frequency. According to solution of formerly proposed 17 DOF vertical vibration model, the time frequency representation of main parts (motor, car and both rope ends) vibration are calculated by short time Fourier transform (STFT) and exhibited in Figure 8.1. Clearly the varying trend of natural frequencies are presented, which helps identify the frequency components.



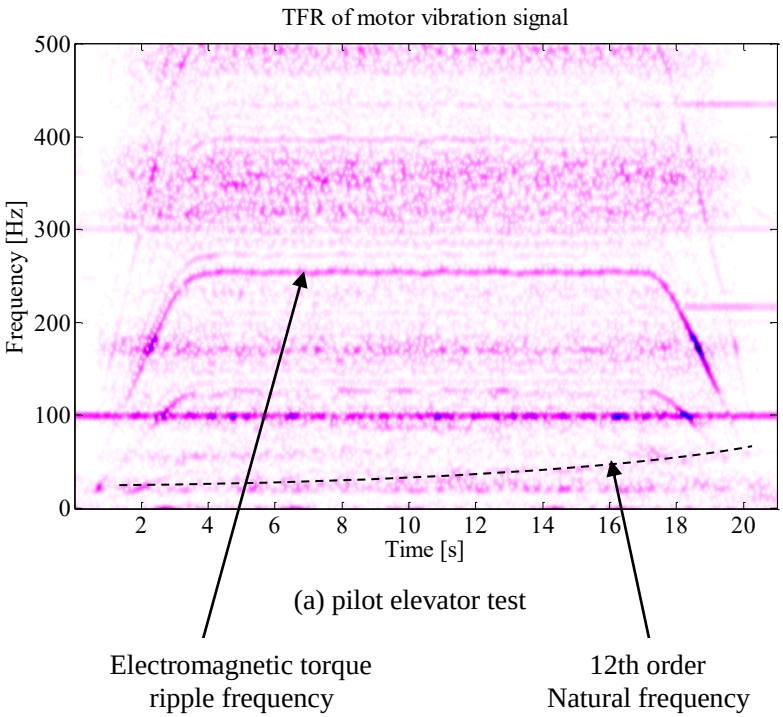

(a) TFR of motor vibration signal (free vibration)



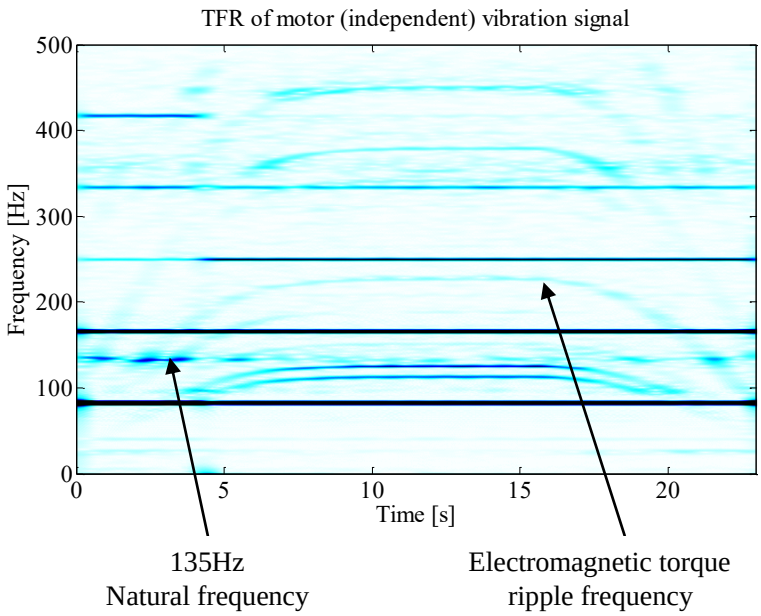

(b) TFR of car vibration signal (free vibration)

Fig 8.1 TFR of vertical free vibration

Time-frequency analysis was also used in analyzing the frequency structure of motor vibration signals. According to the value of characteristic frequencies and corresponding varying trend, we can understand the meaning of main frequency components. Example of TFA's application in motor vibration signal analysis is given in Figure 8.2.




(a) pilot elevator test




(b) lab experimental test

Fig 8.2 TFRs of motor vibration signal

In addition, since car frame vibration signals contain abundant of system running and structural health information, we use time-frequency analysis to reveal the whole time, frequency as well as amplitude information concurrently. TFR of car vibration signal in travel test from 4th floor to 1st floor is exhibited in Figure 8.3, from which the transient impacts, time-variant natural frequency components and characteristic frequency components are legible and recognized.

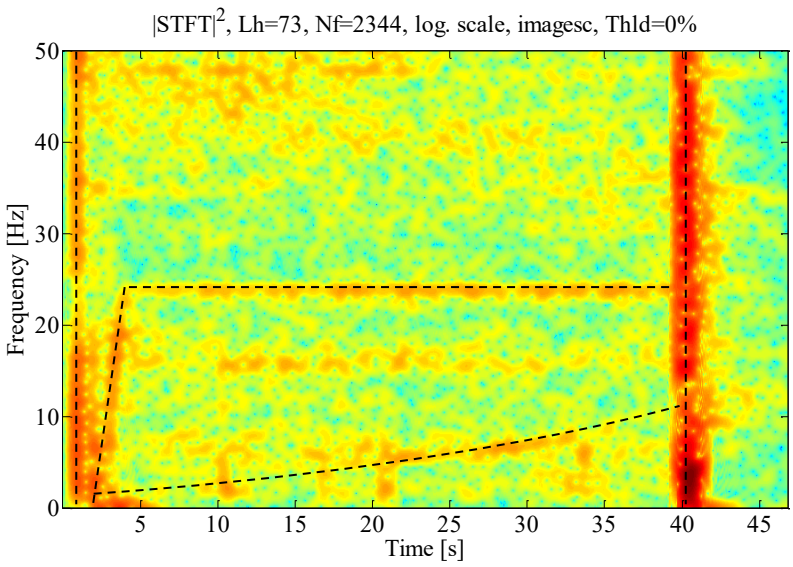



Fig 8.3 TFRs of car frame vibration signal

# 9 CONCLUSIONS

- In first 2 phases of this project much numerical simulated and experimental research have been done to understand the vibration mechanisms of elevator, based on which a series of indicators have been proposed.
- Three frequency domain indicators are specifically proposed for elevator monitoring and diagnosis, especially for rope diagnosis. They were discovered in rope structure study, rope motion and analysis of elevator signals. Formerly the existence of these 3 indicators, as well as their effectiveness, have already been primarily validated.
- For further demonstrating the effectiveness of proposed frequency indicators in real world elevator fault diagnosis, on-site tests on Jinan pilot elevator were carried out, including faulty cases like rope wear, rope local fault and rope loose. Both travel test and modal test/swing test were conducted.
- The comparison of Fourier spectra of pulley vibration signal in travel test between baseline case and rope wear case demonstrate that the rope passing frequency is effective in detecting potential rope wear fault. The properties of rope passing frequency in different wear severity still demands accurate lab experimental research.
- The comparison of Fourier spectra of car frame vibration signal in travel test between baseline case and rope local fault case demonstrate that the rope swing frequency is effective in detecting potential rope local fault.
- The comparison of TFRs of car frame vibration signal in modal test between baseline case and rope loose case indicate that the natural frequency varies in limited scale when rope loose fault is not severe. Hence the sensitivity of system natural frequency still demands accurate lab experimental research.
- The comparison of spectra of car frame vibration signal in rope swing test between baseline case and rope loose case verify the effectiveness of using rope swing frequency to detect potential rope loose fault.

In phase 3 on-site fault experiments were further conducted to verify the effectiveness of former proposals in real-world elevator monitoring and diagnosis. The result exhibited in this report these proposals are meaningful and useful in real applications, fulfilling former expectations.

# REFERENCE

[1] G. Zhang. Study on elevator dynamic characteristic and monitoring & analysis system development. Master Thesis, Zhejiang University, 2008. (in Chinese)

[2] W. Jin, H. Zhang, Y. Zhen. Modeling and dynamic analysis of elevator. Journal of Machine Design and Research, 1999(3): 53-55. (in Chinese)

[3] Y. Wang. Study on elevator vibration attenuation and noise reduction. Master Thesis, Shanghai Jiao Tong University, 2001. (in Chinese)

[4] D. Yu, J. Yu, B. Chen. KED analysis and dynamic performance optimization of express elevator. Chinese journal of applied mechanics, 1998(2): 41-43. (in Chinese)

[5] P. Lei. Research on elevator traction and vibration. Master Thesis, University of Science and Technology Beijing, 2002. (in Chinese)

[6] X. Gang. Dynamic model and parameter optimization method study for traction elevators. Doctoral Thesis, Zhejiang University, 2005. (in Chinese)

[7] J. Zhang, Q. Yang, Y. Wang. Optimal active control of high speed elevator vibration in time domain. Journal of Zhejiang University of Technology, 2000，28(1): 78-83. (in Chinese)

[8] J. Zhang, Q. Yang. Optimal active control of high speed elevator vibration in frequency domain. Journal of Zhejiang University of Technology, 2000, 28(3): 230-235. (in Chinese)

[9] K. Nai, W Forsythe and R M Goodall. Modelling and simulation of a lift system. IMA International conference on control, Sep 1992.

[10] K Nai, W Forsythe and R M Goodall. Improving ride quality in high-speed elevators. World Elevator, 1997, 45(6), 88-93.

[11] Y. Feng. Active Control of Horizontal Vibrations for High-speed Elevator. Doctoral Thesis, Shanghai Jiao Tong University, 2008. (in Chinese)

[12] W. Fu, C. Zhu. Horizontal vibration modeling and dynamic response analysis of high speed elevator. Journal of Machine Design and Research, 2003, 19(6): 65-67. (in Chinese)

[13] L. Guo, G. Zhang, X. Li. Modeling of elevator cabin-rail coupled dynamic system and its properties. Chinese Journal of Mechanical Engineering, 2007, 43(8): 186-190. (in Chinese)

[14] W. Zhu and G. Xu. Vibration of elevator cable with small bending stiffness. Journal of

Sound and Vibration, 2003, 263(3): 679-699.

[15] S. Cao. CIS portable elevator acceleration measurement and analysis system. China elevator, 1998(6): 19-21. (in Chinese)

[16] G. Zhang, Y. Li. Vibration measurement for elevator traction motor, 1999(3): 56-61. (in Chinese)

[17] L. Guo. A study on the measurement and modeling of elevator rail irregularities and the properties of horizontal vibration of cabin. Doctoral Thesis, Tianjin University, 2005. (in Chinese)

[18] Q. Li. Questions on measurement of reconstructive elevator. China elevator, 2000(4): 46-47. (in Chinese)

[19] Y. Chen, S. Shi, et al. A review on fault diagnosis of dynamic systems. Control and Instruments in Chemical Industry, 2001(28): 11-14. (in Chinese)

[20] L. David, R. Jose. Dynamic probabilistic model-based expert system for fault diagnosis. Computers and Chemical Engineering, 2000(24): 2473-2492.

[21] S. Christine, A. Jean, et al. An expert system program for ICP-AES system diagnosis. Spectrochimica Acta Part B, 1997(52): 1923~1927.

[22] J. Zhang, X. Xu, B. Xu. Advanced technology for condition monitoring and fault diagnosis of electric-mechanical systems. Journal of Beijing Institute of Technology, 2004(24): 754-756. (in Chinese)

[23] Z. Tang, J. Bechta. Minimal Cut Set/Sequence Generation for Dynamic Fault Trees. RAMS, 2004: 207-213.

[24] S.K. Ong, N. An. Web-Based Fault Diagnostic and Learning System. International Journal of Advanced Manufacturing Technology, 2001(18): 502-511.

[25] J. Peng. Application of ultrasonic fault detecting technique in escalator detection. China elevator, 1998(7): 26. (in Chinese)

[26] J.C. Harrison. An experimental method for appraisal and comparison of vibration in high-rise elevator cars. Elevator World, 1998, 46(6): 81-85.

[27] Y. Cho, R. Rajamnai. Identification and experimental validation of a scalable elevator vertical dynamic model. Control Engineering Practice, 2001(9): 181-187.

[28] H. Wee, Y. Kim, H. Jung. Nonlinear rate-dependent stick-slip phenomena: modeling and parameter estimation. International Journal of Solids and Structures, 2001(38): 1415-1431.

[29] Z. Zhang, J. Hu. Noise control of high speed elevator. China elevator, 2000(3): 41-43. (in Chinese)

[30] X. Wang. Vibration source analysis of tractor machine makishinko worm reducer. China elevator, 1998(4): 18-19. (in Chinese)

[31] J. Song. Discovery of elevator ride quality factors. China elevator, 1999(1): 43-44. (in Chinese)

[32] F. Yao, Z. Chen. Study on elevator intelligent diagnosis system based multi-sensor

information fusion technology. Master Thesis, Xinjiang University, 2013. (in Chinese)

[33] R. Zhu, Z. Gao. The long-distance monitor and the expert-based fault diagnosis for elevator systems. Master Thesis, Tianjin University, 2003. (in Chinese)

[34] C. Jiang, B. Liu. Research on elevator fault diagnosis based on rough sets. Master Thesis, Xi'an Technology University, 2011. (in Chinese)

[35] M. Duan, B. Liu. Description logic-based intelligent fault diagnosis method. Xi'an Technology University, 2012. (in Chinese)

[36] X. Huang, N. Zhou. Remote monitor center of OTIS Inc. China elevator, 1998(5): 20-22. (in Chinese)

[37] L. Zhang. Elevator development from perspective of interlift'97. China elevator, 1998(6): 10-13. (in Chinese)

[38] H. Chen. Remote condition monitoring system of Kaibo elevator. China elevator, 1999(1): 9-14. (in Chinese)